American International University-Bangladesh (AIUB)

Department of Computer Science

Faculty of Science & Technology (FST)

Cafe Management System

Semester: Fall 25-26

Group: 06		***Section: T***
SL	**Student Name**	**Student ID**
1	JUNAYED ISLAM	23-50150-1
2	SAMIA KHAN	23-50131-1
3	SIFAT AL MUQTADIR	23-50146-1
4	SHARIN BINTE SHAFIQ	23-50525-1
5	RAIYAN YUSUF	23-54376-3

Table of Contents

1. Project Proposal 3
   1.1 Background to the Problem 3
   1.2 Selection of Process Model 3
2. Software Requirements Specification (SRS) / PRD 4
   2.1 Scopes and Features 4
   2.2 User Story Table 7
   2.3 Requirements Traceability Matrix 9
      2.3.1 Functional Requirements 9
      2.3.2 Non-Functional Requirements 11
3. Software Design 14
   3.1 System Design 14
   3.2 UI Design using Figma 17
4. Git Workflow 26
5. Software Testing 30
6. Conclusion 42

1. PROJECT PROPOSAL

1.1 Background to the Problem

Modern cafe's require fast, accurate, and organized service to handle daily operations such as taking orders, preparing food, managing bills, and maintaining records. Many small and medium cafes still depend on manual paper-based processes, which often lead to slow service, calculation errors, missing sales records, and difficulty in tracking inventory or staff activities. These issues reduce customer satisfaction and make business management harder for owners.

To solve these problems, a **Cafe Management System** provides a digital platform where all core operations like order taking, billing, menu management, and user roles are managed in one place. This desktop-based application allows different types of users such as Admin, Staff, Cashier, and Customers to interact with the system according to their roles. The system helps automate repetitive tasks, ensures accuracy, speeds up customer service, and gives the owner full control and visibility over cafe operations.

This project aims to create a structured, user friendly, and efficient system that supports a cafe's day-to-day workflow, minimizes human error, and improves overall business performance.

1.2 Selection of Process Model

For the development of the Cafe Management System, the Extreme Programming (XP) model has been selected. XP is an agile software development methodology that emphasizes high-quality code, continuous feedback, and adaptability to changing requirements, which are essential for this project.

Extreme Programming follows short development iterations, where the system is developed and improved incrementally. Frequent releases ensure that functional and testable components are delivered at regular intervals. This approach allows early detection of errors and ensures that the system continuously aligns with user requirements.

XP strongly focuses on customer involvement, where feedback is obtained frequently throughout the development process. This helps in refining features such as user management, order handling, and administrative controls based on real needs. Practices like Test-Driven Development (TDD), pair programming, and continuous integration help maintain code reliability, reduce defects, and improve overall software quality.

For this project, development activities are organized into small iterations that include planning, coding, testing, and refactoring. This structured yet flexible approach reduces development risk, improves maintainability, and supports rapid changes. Therefore, the XP model is chosen as it provides an efficient and disciplined framework suitable for developing a reliable and scalable Cafe Management System.

2. SOFTWARE REQUIREMENTS SPECIFICATIONS (SRS) / PRODUCT REQUIREMENTS DOCUMENT (PRD)

2.1 Scopes and Features

1. Signup / Register [Customer/Staff]

   - Email/Phone must be unique in the database.

   - Password must be at least 8 characters.

   - "Confirm Password" must match "Password".

   - Successful registration redirects to Login page.

2. Login [All Users]

   - Username and Password cannot be null.

   - System must validate credentials against the database.

   - Successful login redirects to the specific dashboard based on.

   - Invalid credentials show "Incorrect username or password" error.

3. Forgot Password [All Users]

   - System checks if Email exists in the database.

   - If Email exists, an OTP is sent.

   - OTP must match the generated code.

   - New Password must be updated in the database only if OTP is valid.

   - Success message redirects user back to Login.

4. Admin Dashboard [Admin]

   - Add/Edit/Remove Products.

   - Manage Users.

   - View Financial Reports.

- View Profile.

- Edit Profile.

5. Cashier Dashboard [Cashier]

   - Launch POS for new orders.

   - Check past transaction status.

   - Generate end-of-day closing report.

   - View Profile.

   - Edit Profile.

6. Staff Dashboard [Staff]

   - View and update order status.

   - Mark orders as "Served".

   - Manage table availability

   - View Profile

   - Edit Profile.

7. Customer Dashboard [Customer]

   - Customer sees personal account details / View Profile.

   - Browse Menu and place orders.

   - Track status of current food.

   - View past receipts and re-order.

   - Edit profile.

8. Add New Product [Admin]

   - Product Name must be unique.

   - Price must be a positive numeric value.

- "Save" adds the item to the database and updates the Menu view immediately.

9. Remove Product [Admin]

   - Product must exist in database

   - "Save" removes the item from the database and updates the Menu view immediately.

10. Manage Users [Admin]

    - Admin can see all registered Cashiers, Staff, and Customers.

    - Admin cannot delete their own account.

    - Clicking "Add New User" opens a new account for Admin, Cashier, Staff, or Customer.

    - Admin can remove the Cashier, Staff, Customer account.

11. View & Edit Profile [All Users]

    - User details must differ based on the logged-in session.

    - "Role" field is read-only (cannot be changed here).

    - "Edit Profile" button redirects to an update form to change Name/Phone/Password.

12. Order Entry [Cashier]

    - Clicking a product adds it to the Cart list.

    - Increasing quantity automatically recalculates the Total Amount.

    - Place Order assigns a unique Order ID.

    - System must check stock availability.

13. Payment Process [Cashier]

    - Cash Received must be greater than or equal to Total Amount.

    - Print Receipt generates a PDF or sends a command to the thermal printer.

14. Order View [Staff]

    - Orders placed by Customer appear here instantly.

    - Staff can toggle status: Pending > Cooking > Ready > Served.

15. Menu View [Customer]

    - Only products marked as available and added by Admin.

    - Interface should be read-only.

16. Generate Receipt [Cashier]

    - Receipt must contain: Date, Time, Order ID, Name.

    - Tax calculation logic must be applied to the subtotal.

17. Logout [All Users]

    Clicking logout clears the session data.

    - User is redirected back to the Login Screen.

2.2 User Story Table

As a/an	I want to	So that	Acceptance Criteria
Admin	Create staff account	I can add new employees to the system	Staff registration form → success message after saving
	Create cashier account	Cashiers can manage payments	Cashier added → visible in cashier list
	Create customer profile (optional)	I can help customers register	Customer receives confirmation message
	Login with username & password	The system can authenticate me	Successful login redirects to Admin Dashboard
	View my profile	I can check my information	Profile page shows all registered data
	Update my information	I can correct wrong details	Updated data instantly reflected in the system
	Change password	My account stays secure	Old password + new password validation + success alert

	Remove staff/cashier	Unwanted users lose access	Confirmation popup → user removed from system
	Add new food item	Customers can order the item	New item saved + visible in menu list
	Update food information	Everyone sees correct data	Updated menu list is shown
	Remove food item	It no longer appears on menu	Food list updates without that item
	View all menu items	I can monitor availability	Table list with Edit & Delete options
	View sales report	I can track profit	Monthly/Yearly report table visible
Staff	Login to my account	I can access my dashboard	Valid credentials → dashboard shown
	View my profile	I can check my data	Profile details visible
	Update my information	I can correct mistakes	Data updated successfully
	Change password	My account is safe	Old + new password required
	View customer orders	I can prepare food	Pending orders list displayed
	Update order to "Preparing"	Admin/Cashier can track progress	Status changes successfully
	Update order to "Ready"	Cashier can deliver food	Status updated to READY
	View menu list	I know item availability	Menu table displayed
Customer	Create an account	I can access cafe services	Confirmation message after registration
	Login with username & password	I can access my profile	Successful Login → Customer Dashboard
	View my Profile	I can see my data	Profile page displayed
	Update my Information	I can fix wrong info	Update success
	Change password	My account stays secure	Password changed
	View menu items	I can choose food	Menu list visible
	Search for food item	I can find items faster	Search results shown
	Add Food to cart	I can select item	Cart updates
	Place an order	I can buy food	Order saved (Pending)
	Make Payment	I can complete my order	Status (Paid)
	View Order Status	I can track my order	Shows Order status
	View Order History	I can see past purchases	History table shown

Cashier	Login with username & password	I can access my dashboard	Successful login > Cashier dashboard
	View my profile	I can see my information	Profile list with Edit button
	Update my information	I can keep accurate details	Data updated successfully
	Change password	My account stays protected	Old + new password validation
	View all pending orders	I can manage payment collection	Pending orders shown with details
	Confirm payment	Customer can receive food	Order status > Paid
	View all paid orders	I can monitor completed transactions	Table showing all paid orders
	Generate bill	Customer can receive food	Printable bill displayed
	View daily sales report	I can track daily income	Table of total amount collected

2.3 Requirements Traceability Matrix

2.3.1 Functional Requirements

Functional requirements define the core features and capabilities of the Cafe Management System. These requirements focus on user interactions, system behavior, and the services provided to different user roles such as Admin, Cashier, Staff, and Customer.

i. User Registration and Authentication

The system supports a structured registration process that allows both customers and staff members to create accounts. During registration, users are required to provide necessary personal and login information, ensuring unique identification within the system. This process enables controlled access and ensures that each user interacts with the system according to their assigned role.

Authentication is a fundamental feature of the system. Users are required to log in using valid credentials before accessing system functionalities. This ensures that only authorized users can access sensitive features such as order processing, payment handling, and administrative controls.

To enhance usability, the system includes a password recovery mechanism. Users who forget their login credentials can securely reset their password using a recovery process.

This reduces account lockouts and improves overall user satisfaction while maintaining security.

ii. Role-Based Access and Dashboards

The Cafe Management System is designed with role-based access control to ensure that users can only access features relevant to their responsibilities. Each role—Admin, Cashier, Staff, and Customer—has a dedicated dashboard tailored to their tasks.

Admins can access administrative tools such as product management and user control.

Cashiers can manage orders, process payments, and generate receipts.

Staff members can view and track incoming orders.

Customers can browse the menu and interact with their account information.

This role-based structure improves efficiency, reduces errors, and enhances system security by restricting unauthorized access.

iii. Product and Menu Management

The system provides comprehensive menu management capabilities for administrators. Admin users can add new products, update existing product details, or remove items from the menu. Each product includes attributes such as name, price, category, and availability status.

This functionality ensures that the menu remains accurate and up to date, reflecting real-time changes in pricing or product availability. Proper product management directly impacts customer satisfaction and order accuracy.

iv. User Account Management

Administrative users have the ability to manage user accounts within the system. This includes viewing registered users, assigning roles, updating user information, and disabling accounts when necessary. This feature ensures smooth operational control and proper system governance.

All users, regardless of role, are allowed to view and edit their own profile information. This includes updating personal details such as name, contact information, and password. Allowing users to manage their profiles improves data accuracy and reduces administrative workload.

v. Order Processing and Payment Handling

The system supports a structured order entry process handled by the cashier. Cashiers can select products from the menu, specify quantities, and create customer orders efficiently. This process minimizes manual errors and speeds up service during peak hours.

Once an order is created, the system enables secure payment processing. The cashier can complete transactions using supported payment methods while ensuring that payment data is handled safely and accurately. Successful payment confirmation updates the order status automatically.

After payment completion, the system generates a receipt containing order details, pricing, and transaction information. This receipt can be printed or displayed digitally, serving as proof of purchase for customers and records for the café.

vi. Order Viewing and Menu Browsing

Staff members have access to order details to support order preparation and delivery. This ensures that staff can efficiently view pending and completed orders, reducing miscommunication between service roles.

Customers can browse the menu through the system interface. The menu display includes product names, prices, and categories, allowing customers to make informed purchasing decisions. This feature enhances transparency and improves the overall customer experience.

vii. Session Management

The system allows all users to securely log out after completing their tasks. Proper session termination prevents unauthorized access, especially when the system is used on shared devices. Logout functionality ensures that user sessions are properly closed and system security is maintained.

2.3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes of the system. These requirements ensure that the Cafe Management System operates efficiently, securely, and reliably under various conditions.

i. Performance Requirements

The system is designed to provide fast and responsive interactions. Under normal operating conditions, user actions such as login, menu loading, and order submission are processed promptly to avoid delays. The system is capable of handling multiple concurrent users without noticeable performance degradation, ensuring smooth operation during peak business hours.

ii. Security Requirements

Security is a critical aspect of the system. Access to system features is controlled through role-based permissions, ensuring users can only perform actions authorized for their role.

User credentials are protected using encryption techniques to prevent unauthorized access and data breaches. Passwords are never stored in plain text, ensuring compliance with standard security practices.

To reduce the risk of unauthorized access, the system includes an automatic session timeout mechanism. Users who remain inactive for a predefined period are logged out automatically, protecting the system from misuse.

iii. Usability Requirements

The system emphasizes ease of use and accessibility. The user interface is designed to be intuitive, allowing users with minimal technical knowledge to navigate the system efficiently.

The application is accessible through standard web browsers, eliminating the need for specialized software installations. This ensures compatibility across different devices and operating systems commonly used in café environments.

iv. Reliability Requirements

The system ensures reliable operation during order and payment processing. Data consistency is maintained to prevent issues such as duplicate orders or incorrect billing.

In the event of unexpected failures such as power loss or system crashes, the system is designed to prevent data loss. Proper error handling and data management mechanisms ensure that critical information remains intact.

v. Maintainability and Scalability Requirements

The Cafe Management System follows a modular design approach, allowing individual components to be updated or modified without affecting the entire system. This simplifies maintenance and reduces downtime during updates.

The system architecture supports future enhancements, such as adding new features or integrating additional services, without requiring major redesign. This ensures long-term usability and adaptability as business needs evolve.

vi. Availability Requirements

The system is expected to remain available during standard business hours with minimal downtime. Regular maintenance activities should be planned in a way that does not disrupt normal café operations. High availability ensures that users can place orders, process payments, and manage operations without interruption.

In situations where system maintenance or updates are required, the system should notify users appropriately and ensure a smooth transition back to normal operation once maintenance is complete.

vii. Data Integrity Requirements

The system ensures that all data entered, processed, and stored remains accurate and consistent throughout its lifecycle. Validation mechanisms are implemented to prevent incorrect, incomplete, or duplicate data entries, particularly during order placement and payment processing.

Any updates to critical data, such as product pricing or user roles, are handled in a controlled manner to prevent inconsistencies across the system.

3. SOFTWARE DESIGN

3.1 System Design:

- Use Case Diagram:

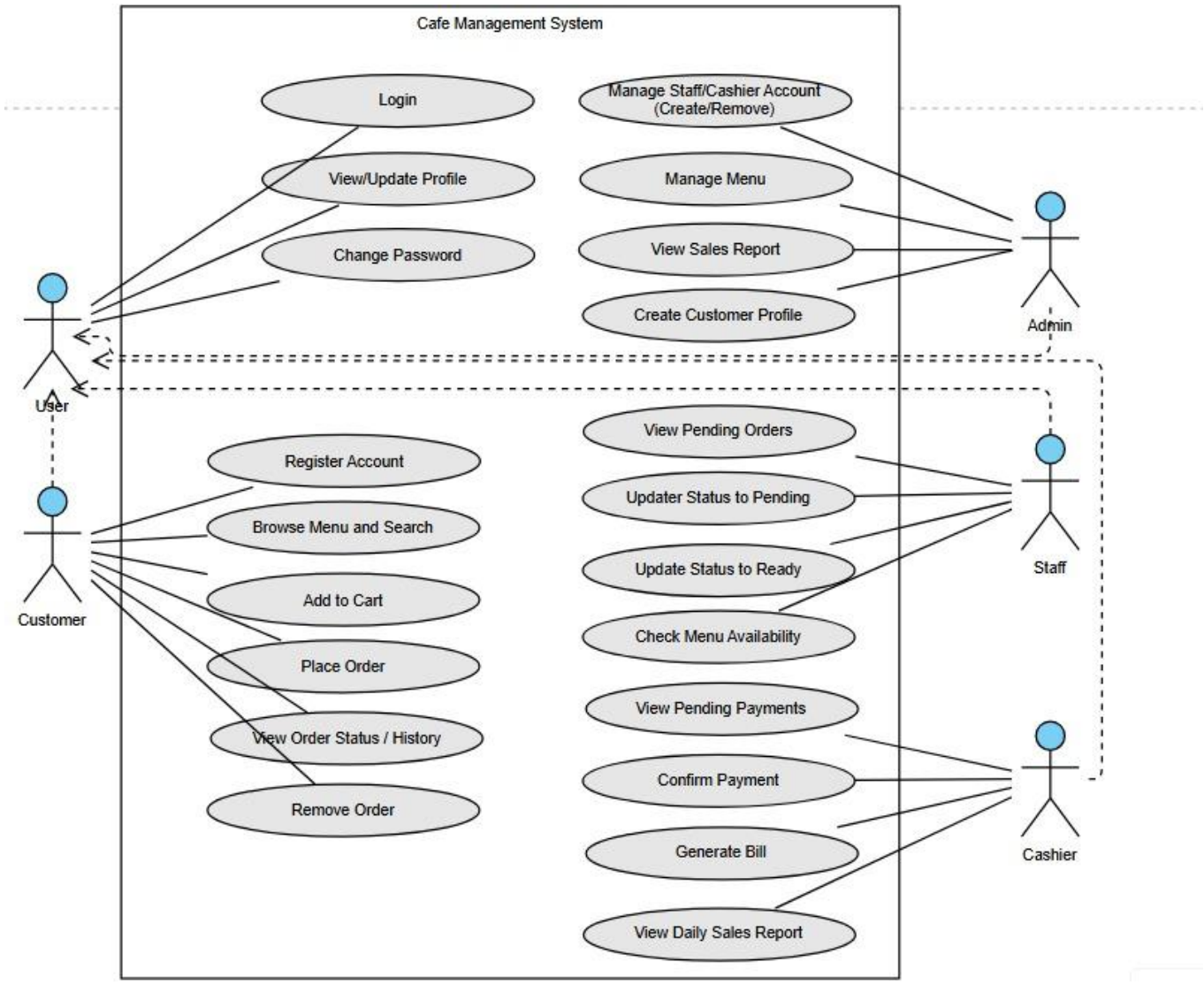


Fig: Use Case Diagram

- Class Diagram

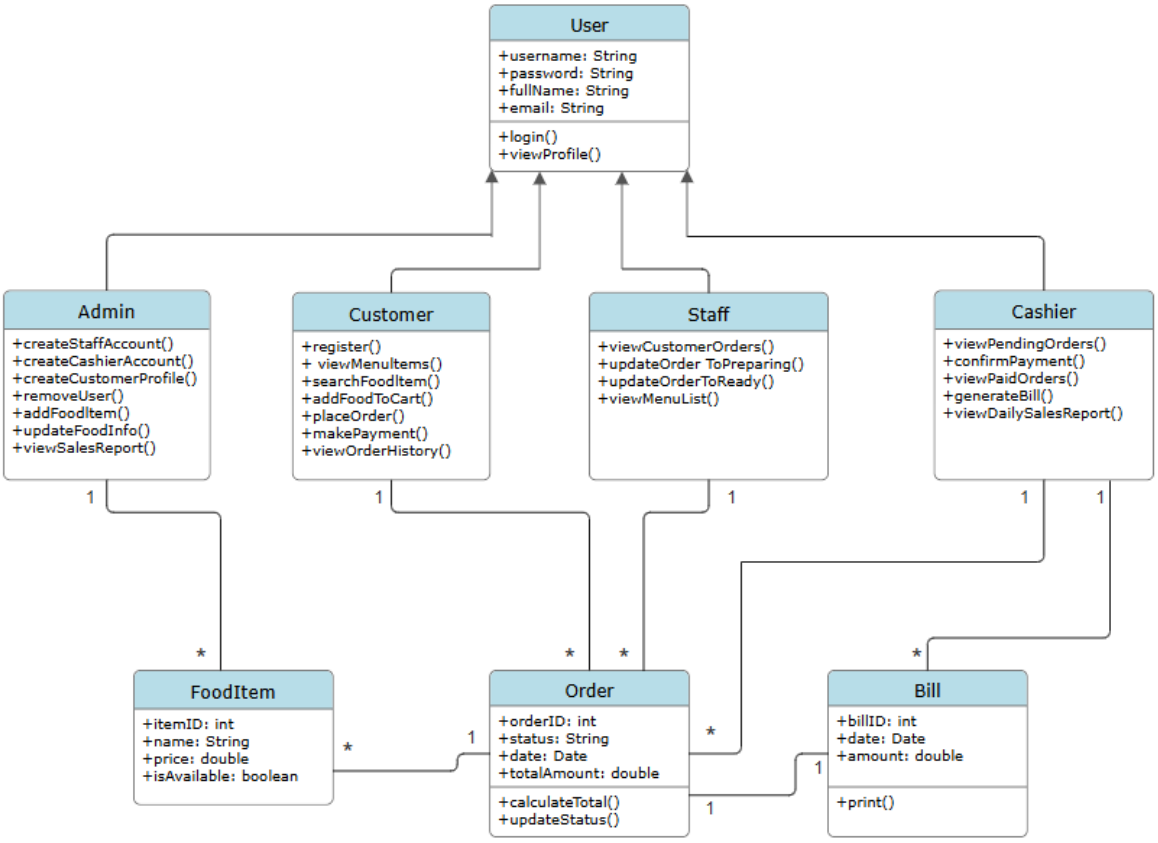


Fig: Class Diagram

- Activity Diagram:

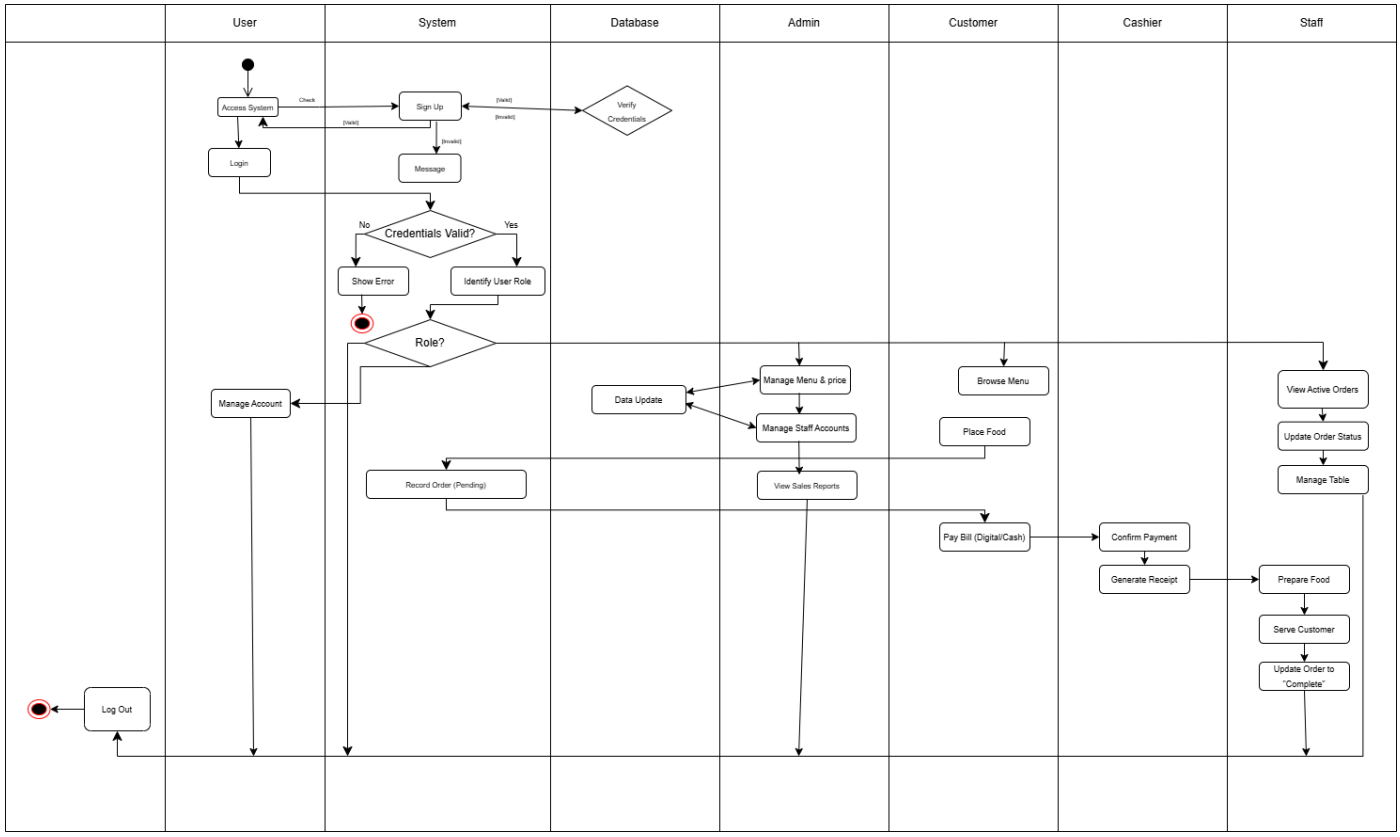
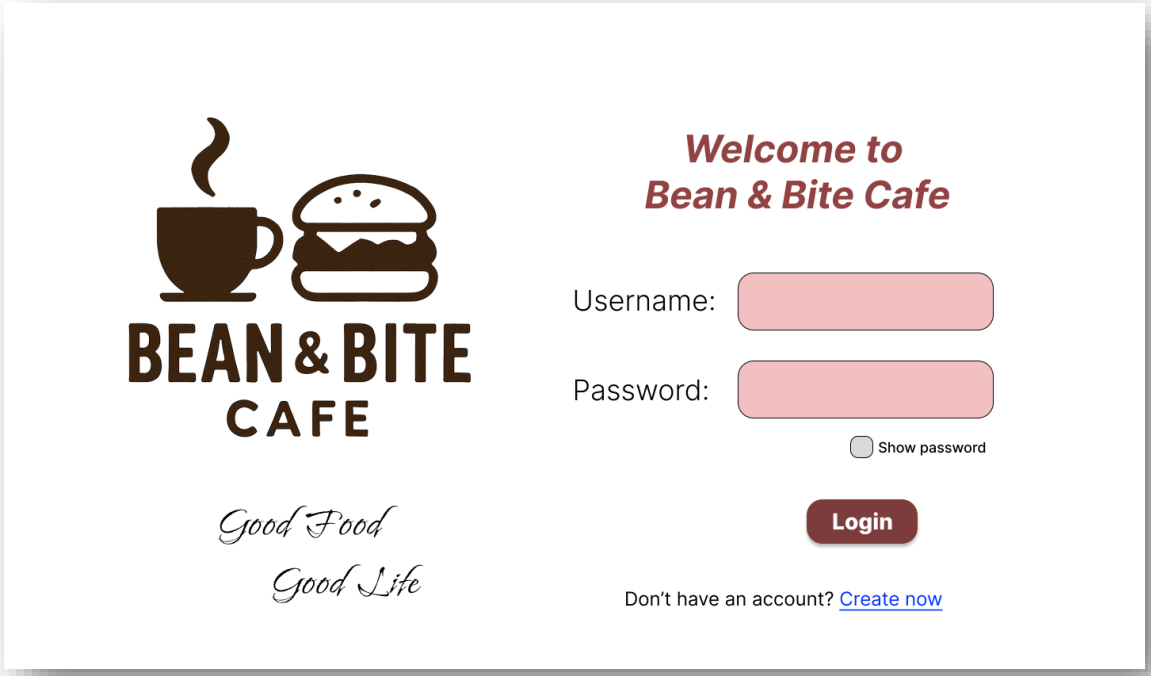
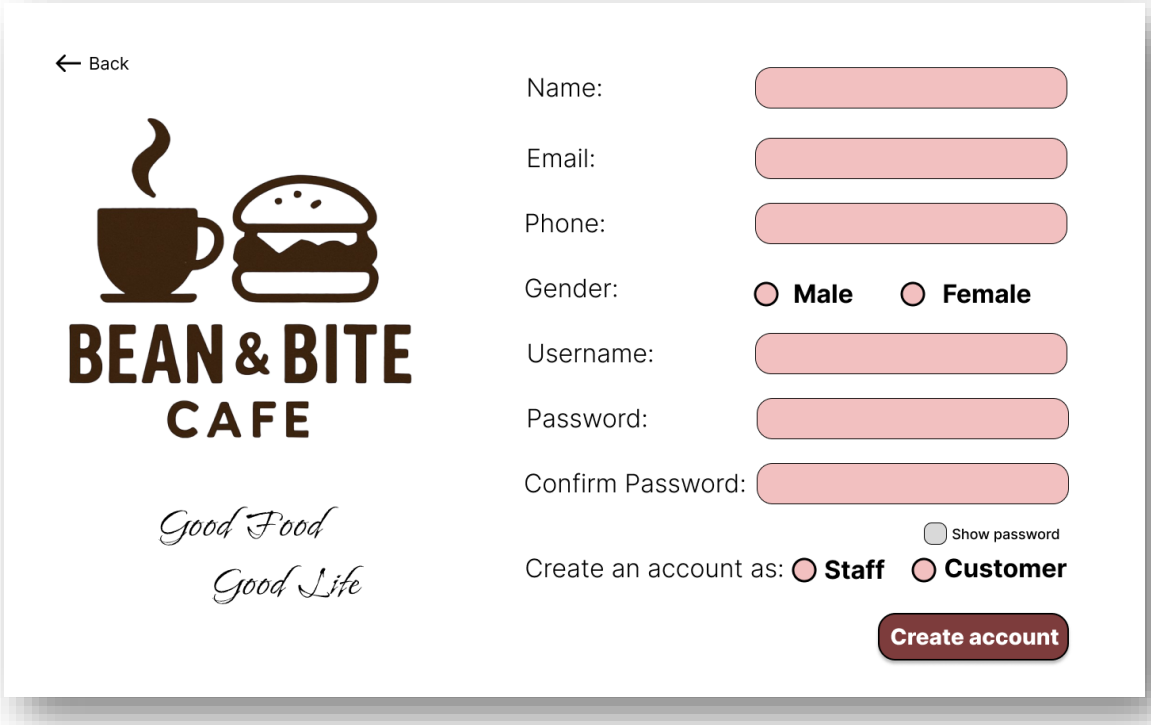
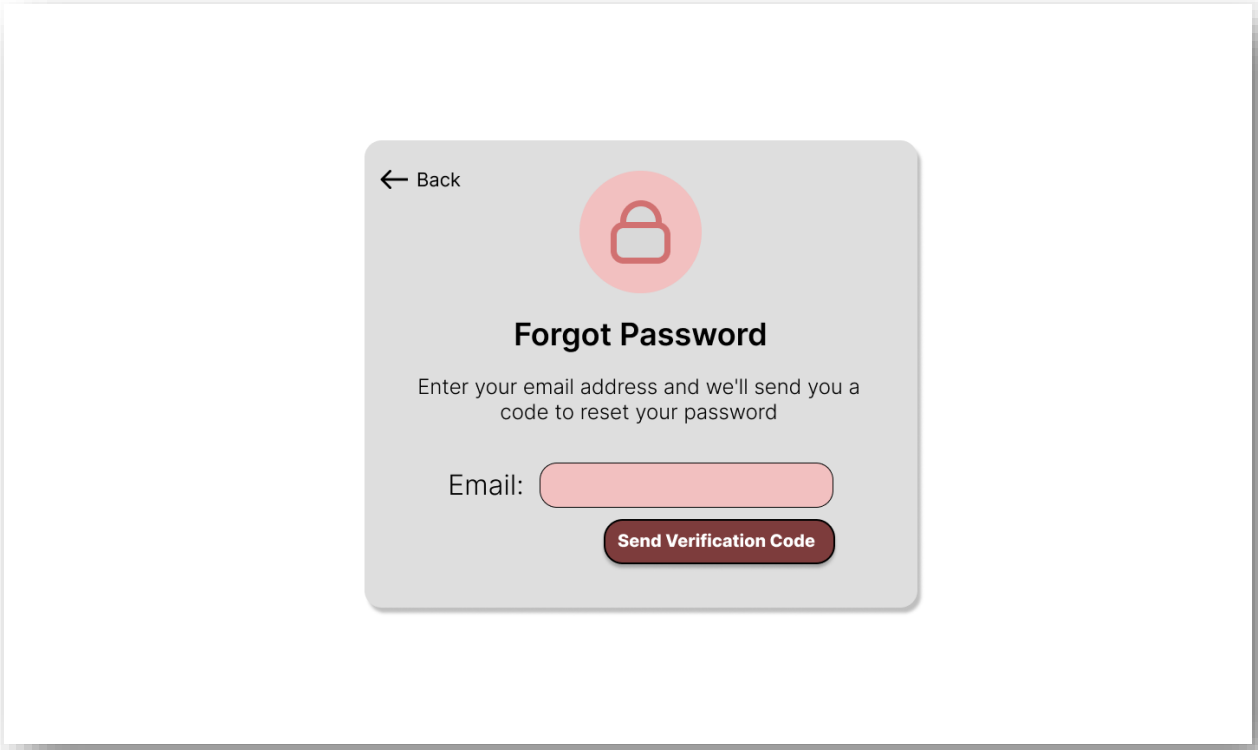


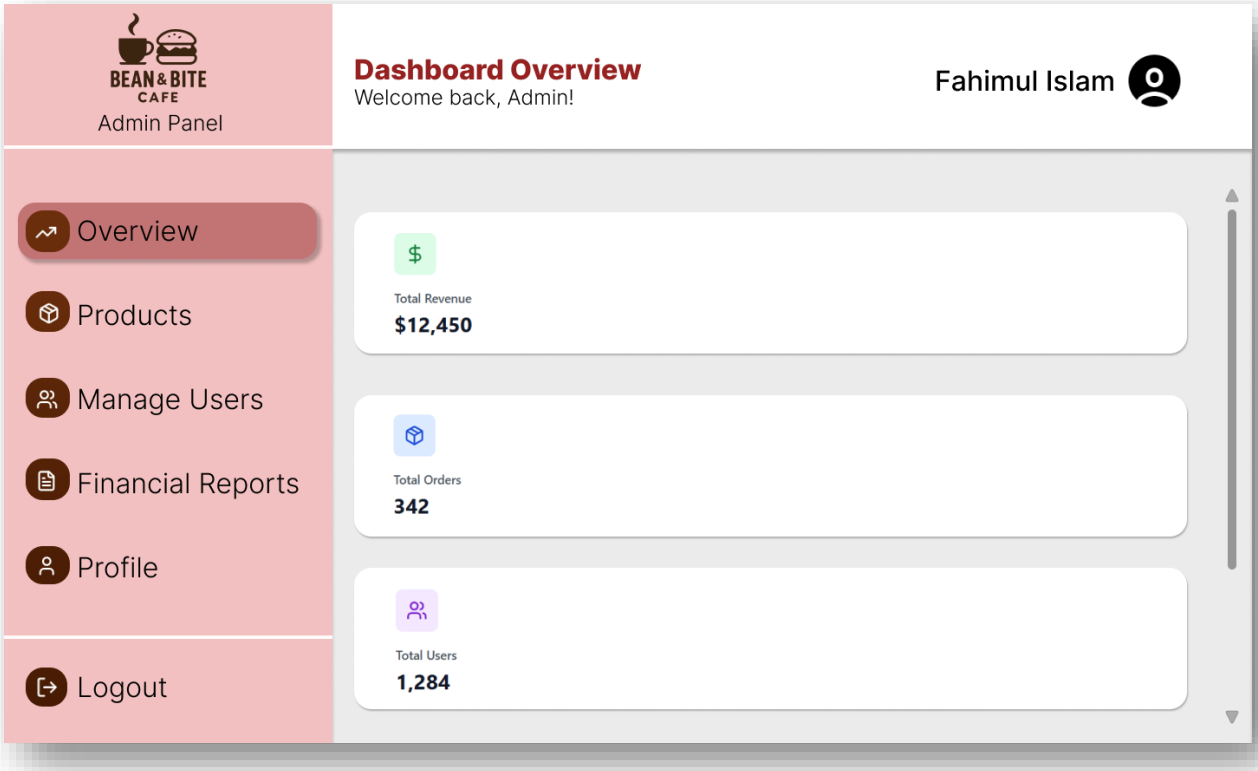
Fig: Activity Diagram

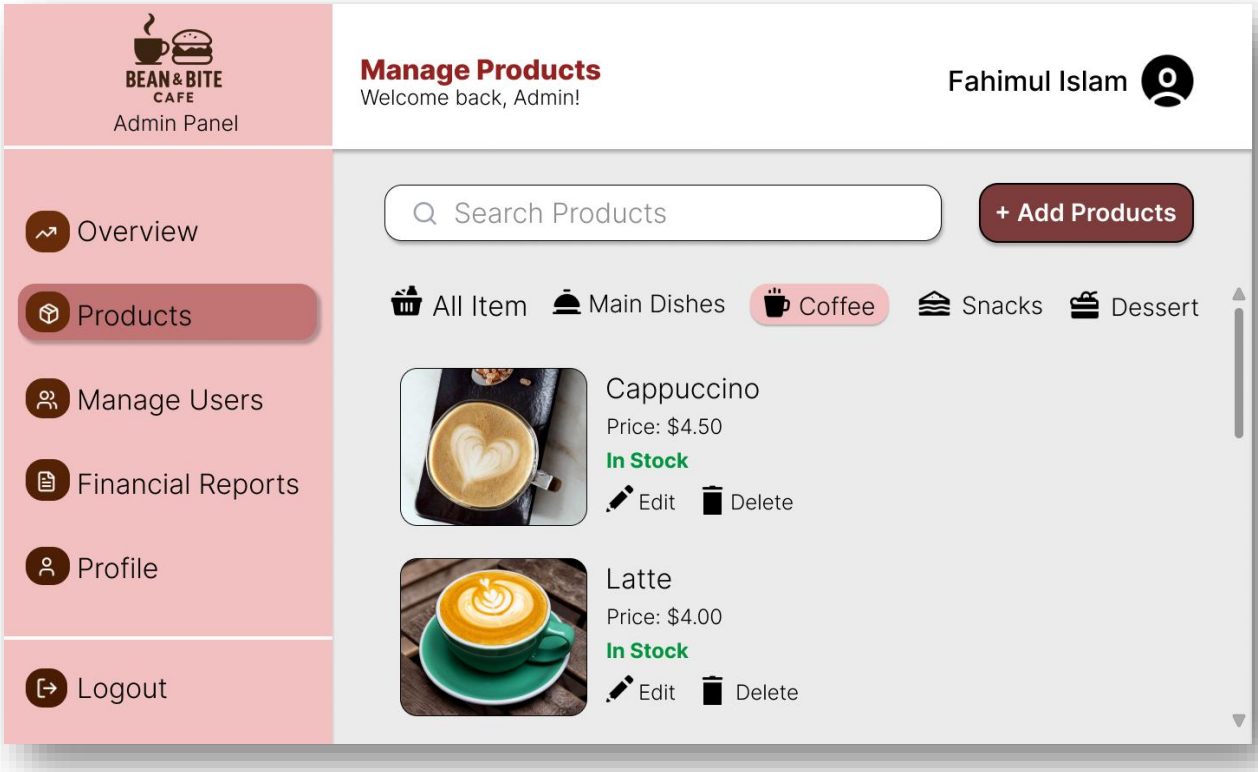
3.2 UI / Wireframe Design using Figma

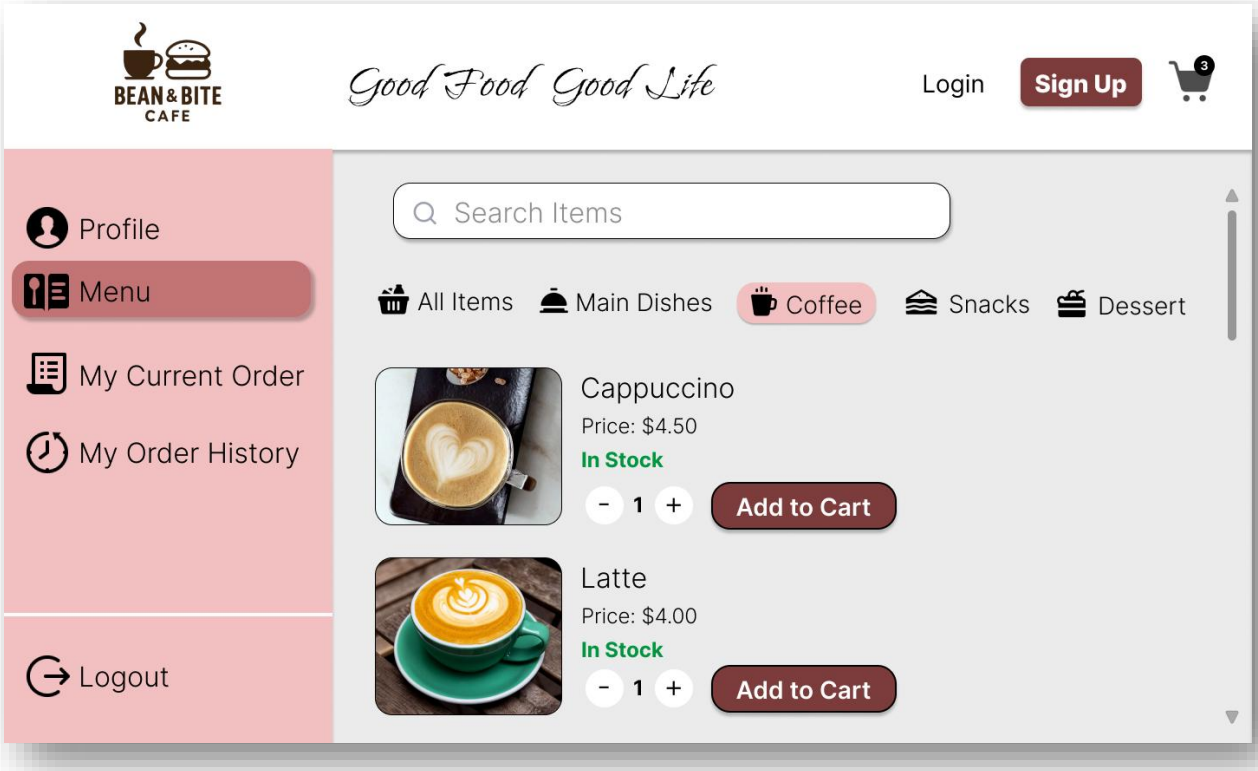


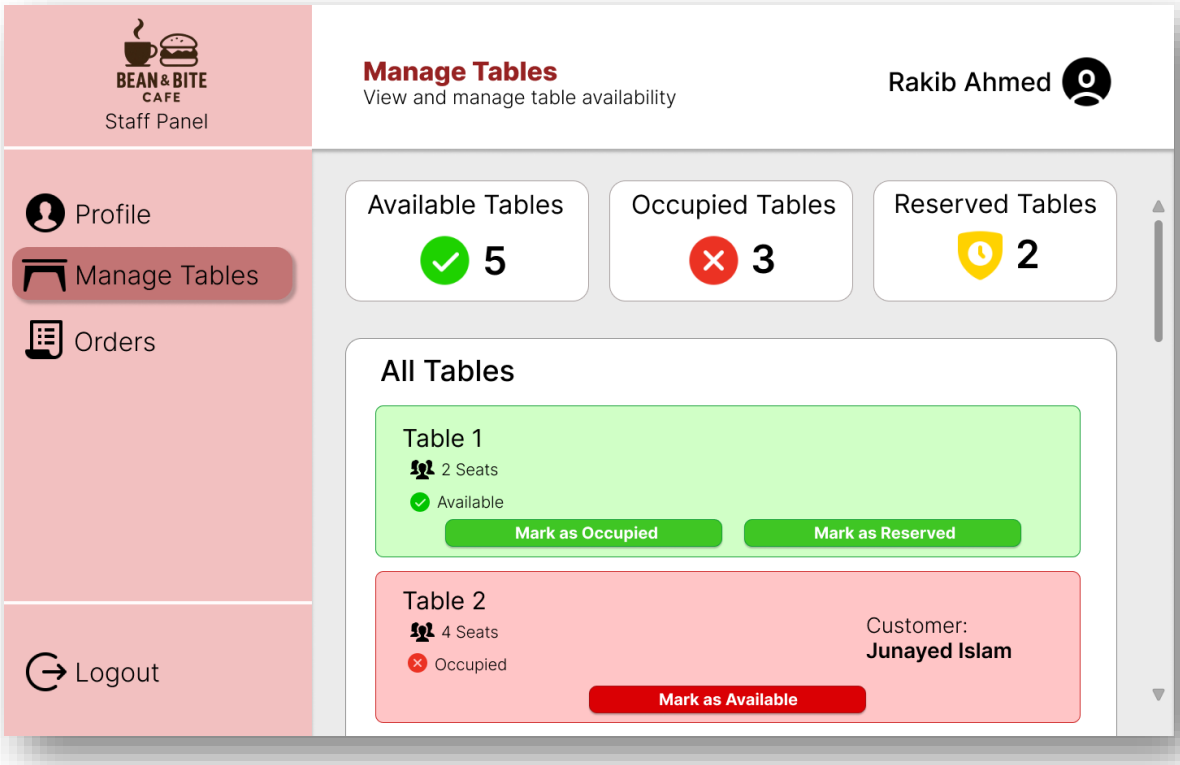


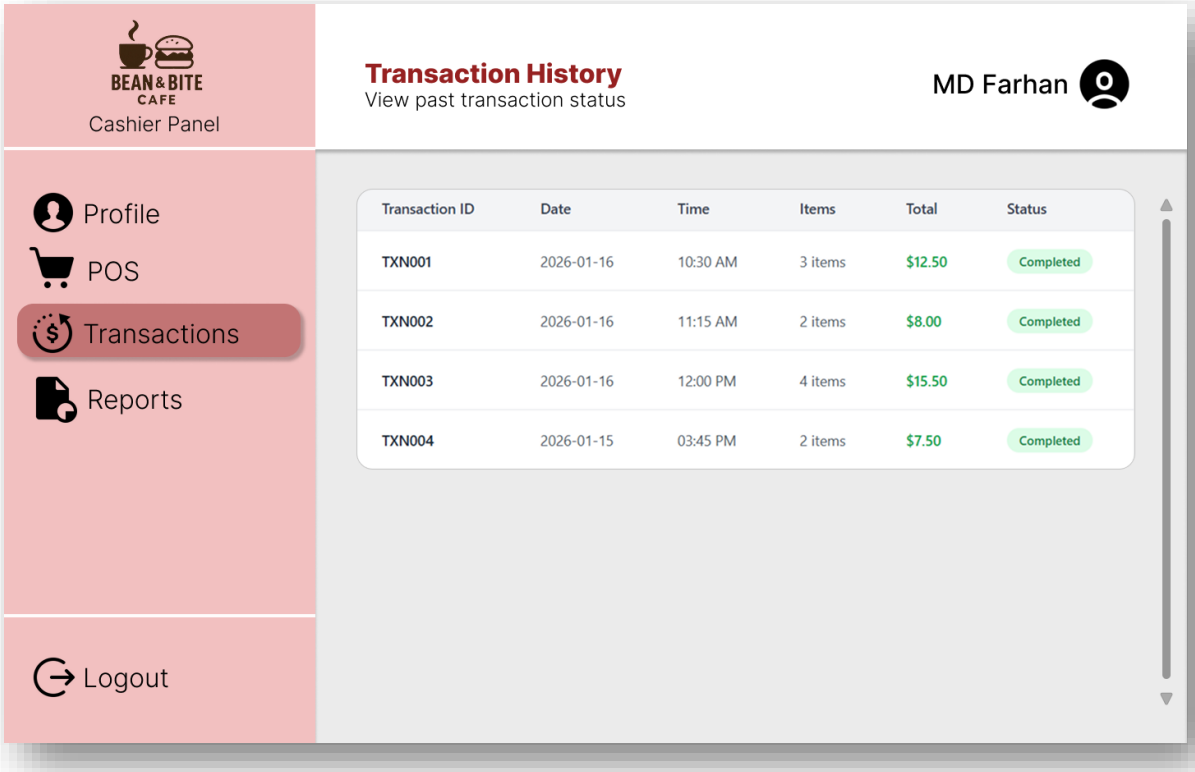

Back
Forgot Password
Enter your email address and we'll send you a code to reset your password
Email:
Send Verification Code

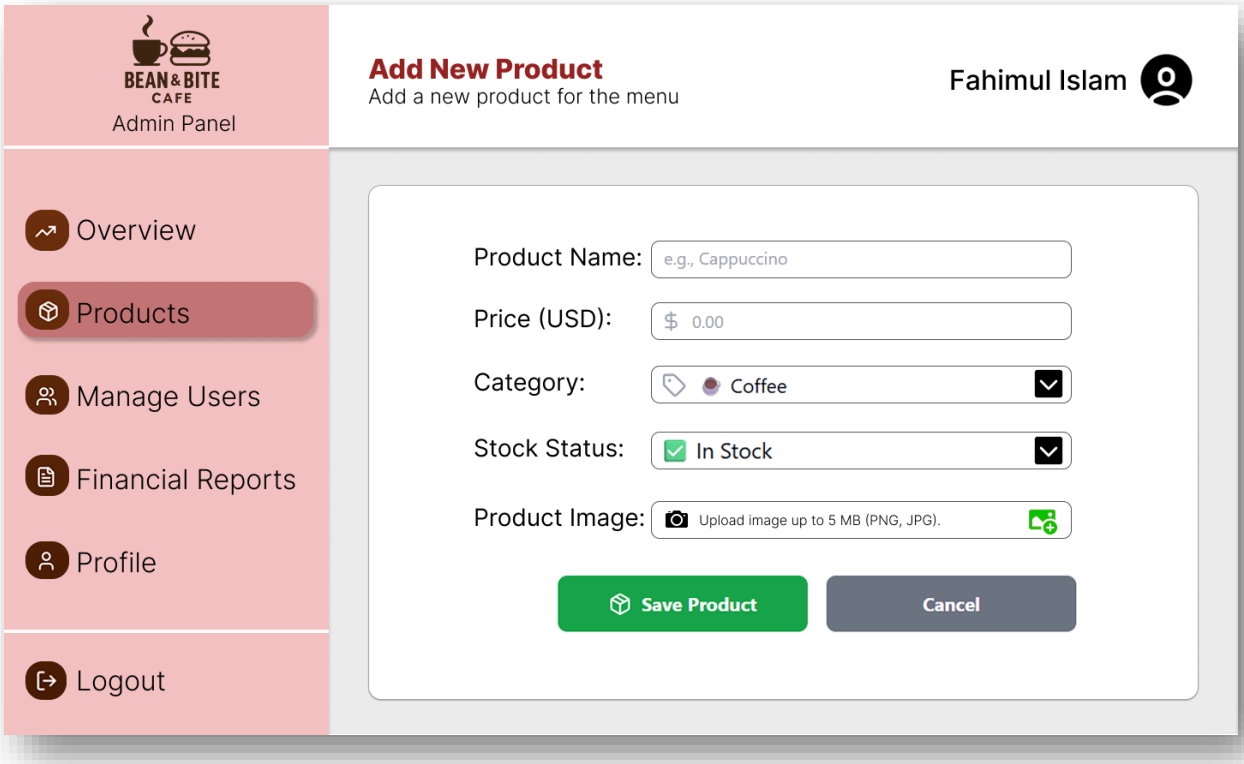

BEAN & BITE CAFE
Admin Panel
Dashboard Overview
Welcome back, Admin!
Fahimul Islam
Overview
Products
Manage Users
Financial Reports
Profile
Logout
Total Revenue
$12,450
Total Orders
342
Total Users
1,284

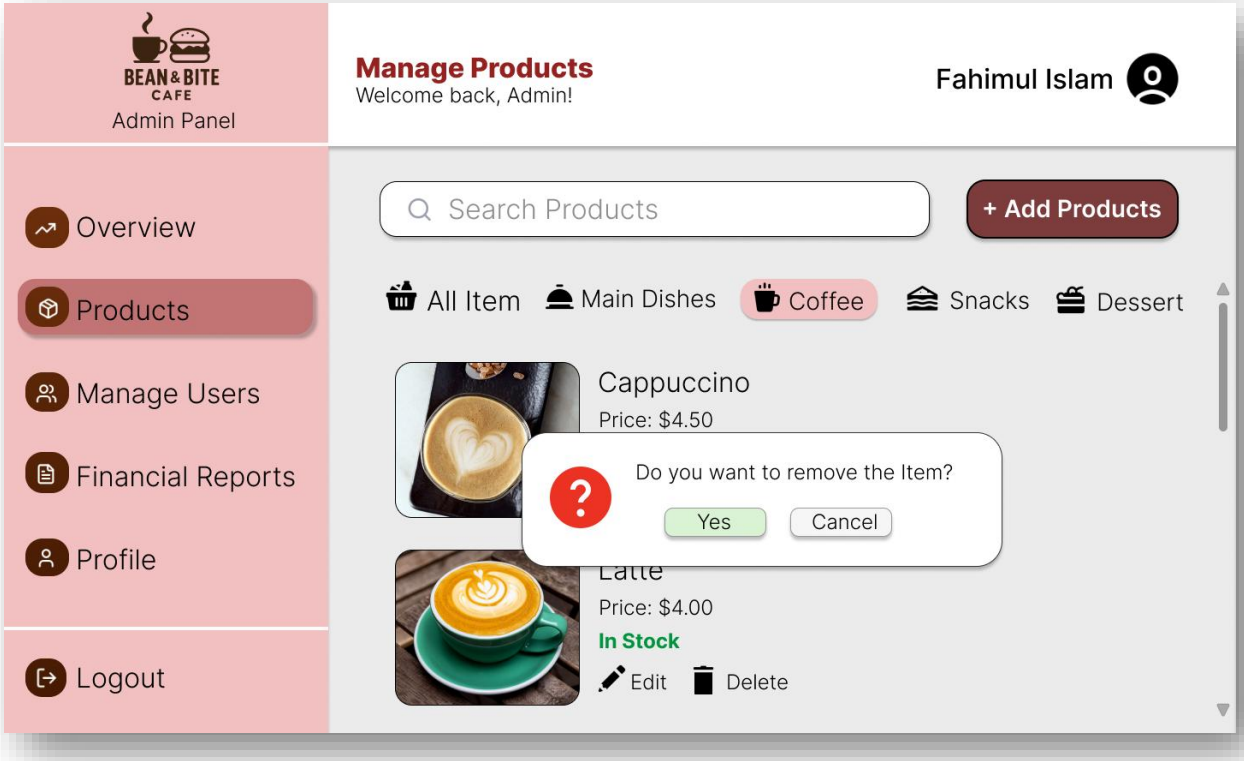

BEAN & BITE
CAFE
Admin Panel
Manage Products
Welcome back, Admin!
Fahimul Islam
Overview
Products
Manage Users
Financial Reports
Profile
Logout
Search Products
+ Add Products
All Item
Main Dishes
Coffee
Snacks
Dessert
Cappuccino
Price: $4.50
In Stock
Edit
Delete
Latte
Price: $4.00
In Stock
Edit
Delete

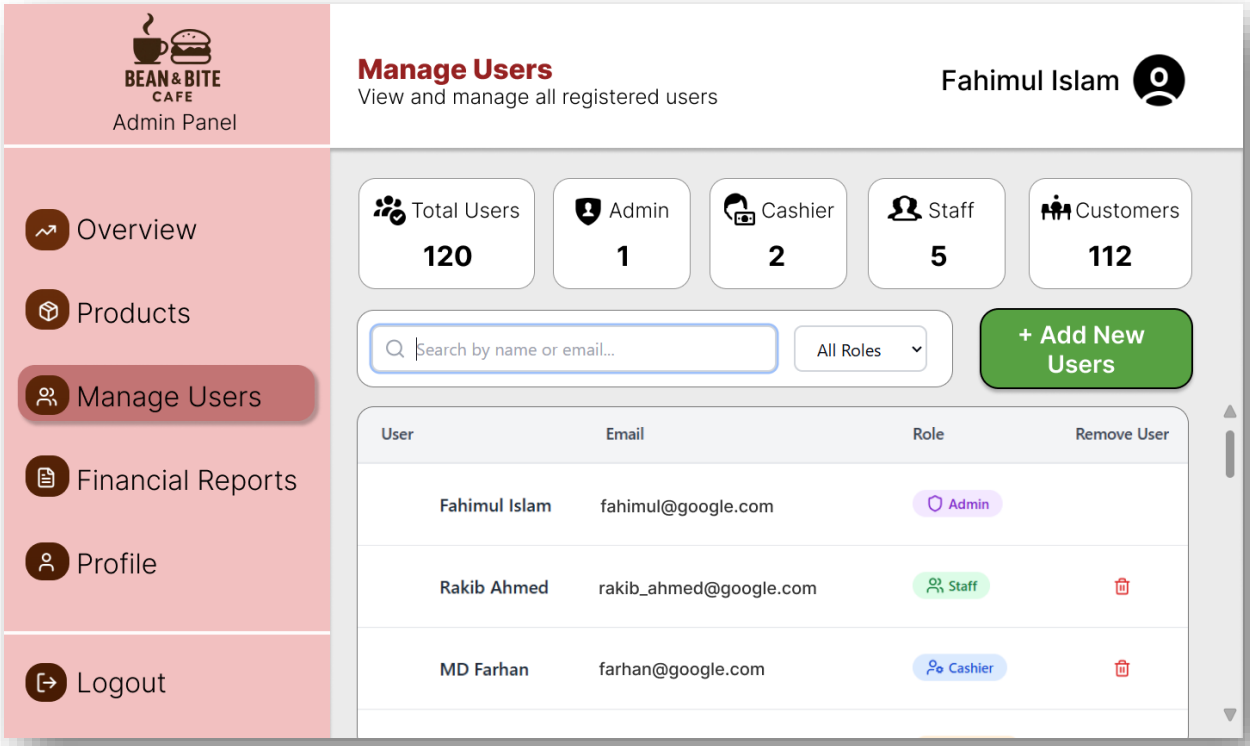

BEAN & BITE
CAFE
Good Food Good Life
Login
Sign Up
3
Profile
Menu
My Current Order
My Order History
Logout
Search Items
All Items
Main Dishes
Coffee
Snacks
Dessert
Cappuccino
Price: $4.50
In Stock
- 1 +
Add to Cart
Latte
Price: $4.00
In Stock
- 1 +
Add to Cart

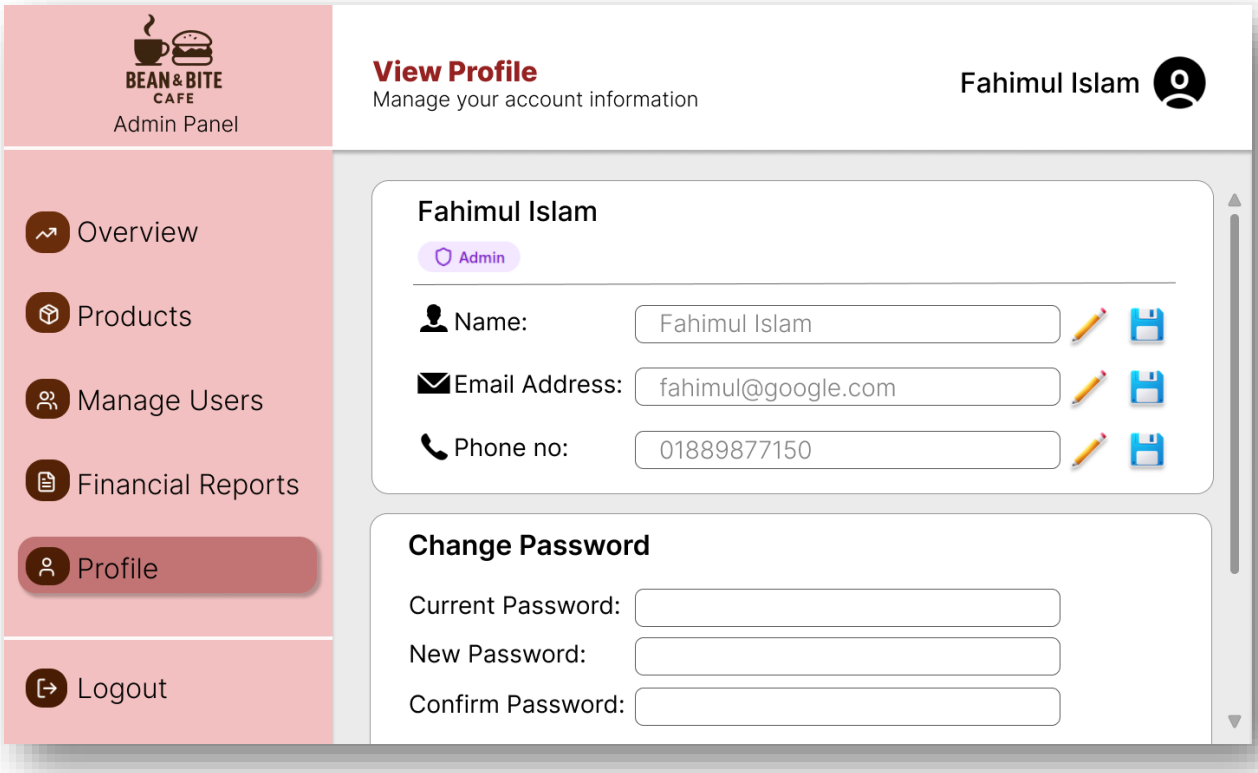

BEAN & BITE
CAFE
Staff Panel
Manage Tables
View and manage table availability
Rakib Ahmed
Profile
Manage Tables
Orders
Logout
Available Tables
5
Occupied Tables
3
Reserved Tables
2
All Tables
Table 1
2 Seats
Available
Mark as Occupied
Mark as Reserved
Table 2
4 Seats
Occupied
Customer:
Junayed Islam
Mark as Available

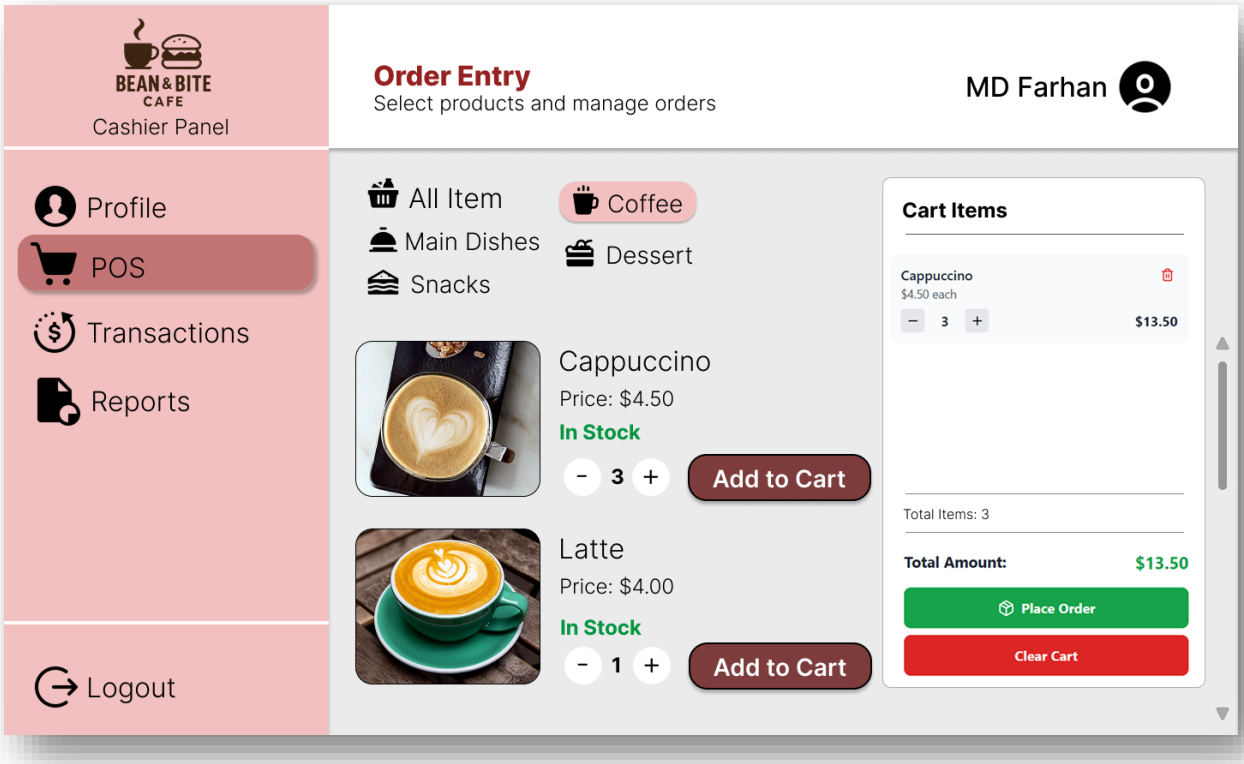

BEAN & BITE
CAFE
Cashier Panel
Transaction History
View past transaction status
MD Farhan
Profile
POS
Transactions
Reports
Logout
Transaction ID
Date
Time
Items
Total
Status
TXN001
2026-01-16
10:30 AM
3 items
$12.50
Completed
TXN002
2026-01-16
11:15 AM
2 items
$8.00
Completed
TXN003
2026-01-16
12:00 PM
4 items
$15.50
Completed
TXN004
2026-01-15
03:45 PM
2 items
$7.50
Completed

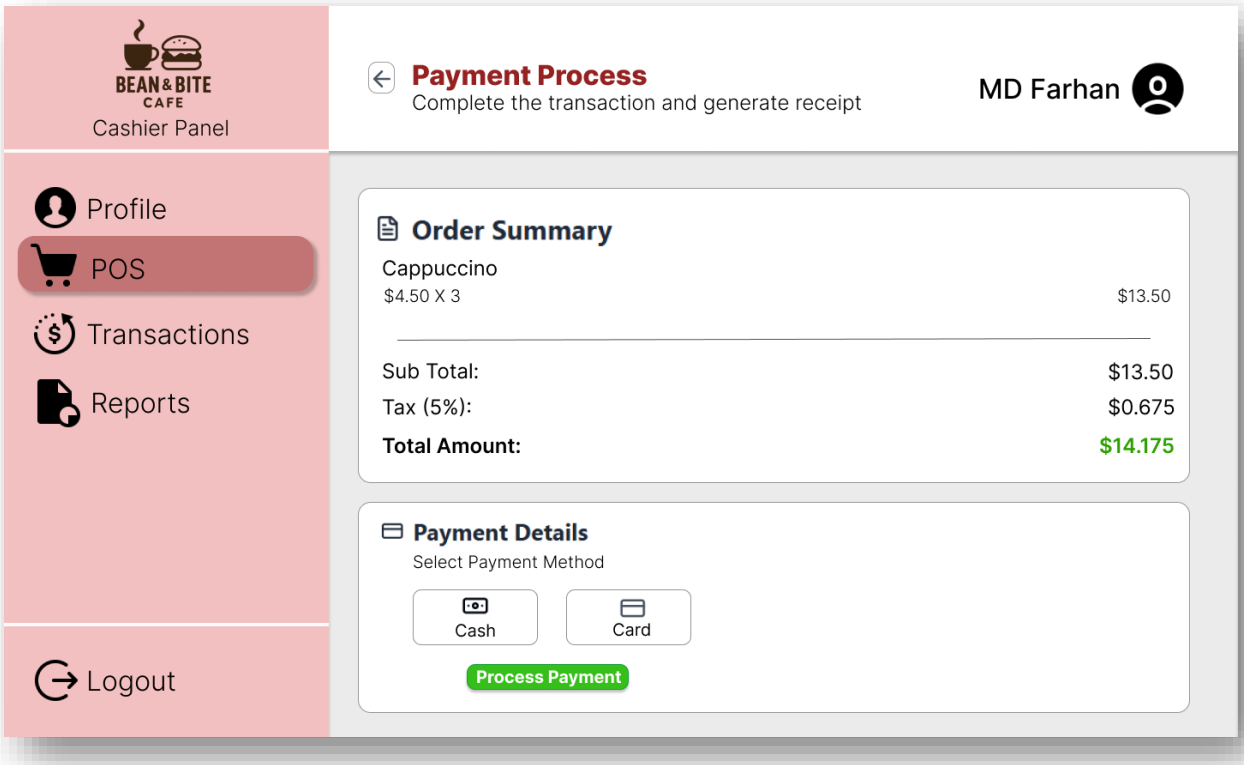

BEAN & BITE
CAFE
Admin Panel
Add New Product
Add a new product for the menu
Fahimul Islam
Overview
Products
Manage Users
Financial Reports
Profile
Logout
Product Name:
e.g., Cappuccino
Price (USD):
$ 0.00
Category:
Coffee
Stock Status:
In Stock
Product Image:
Upload image up to 5 MB (PNG, JPG).
Save Product
Cancel

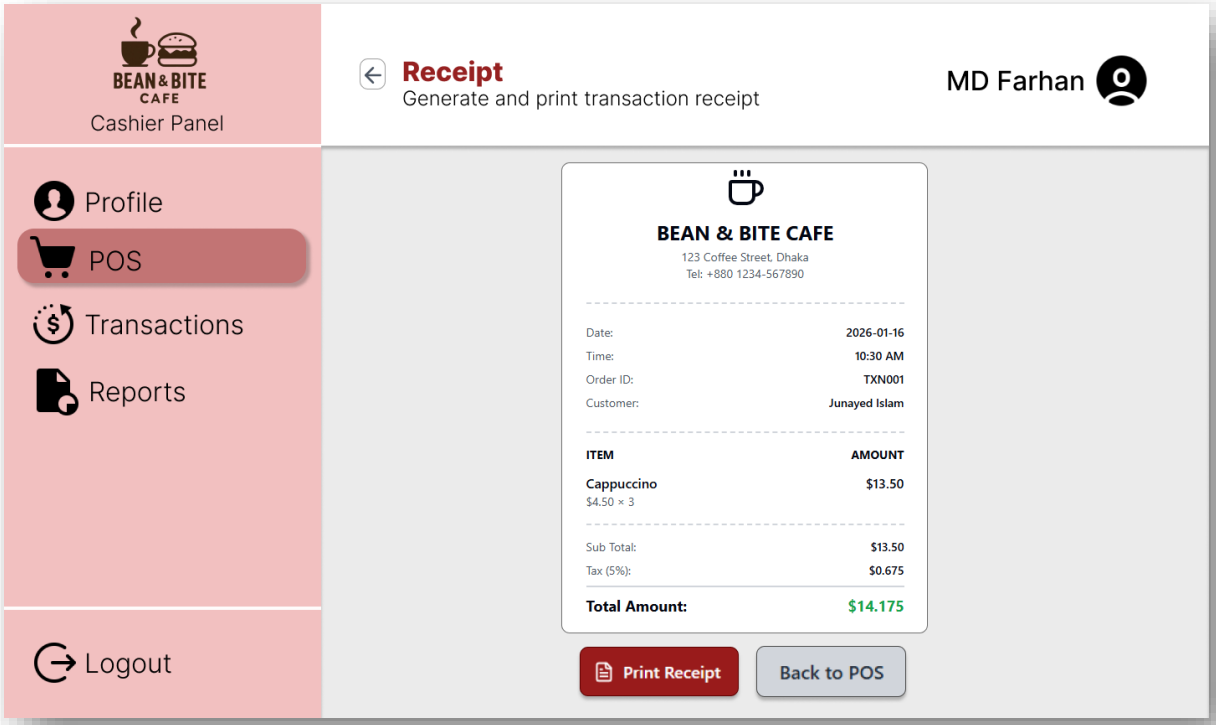

BEAN & BITE
CAFE
Admin Panel
Manage Products
Welcome back, Admin!
Fahimul Islam
Overview
Products
Manage Users
Financial Reports
Profile
Logout
Search Products
+ Add Products
All Item
Main Dishes
Coffee
Snacks
Dessert
Cappuccino
Price: $4.50
Do you want to remove the Item?
Yes
Cancel
Latte
Price: $4.00
In Stock
Edit
Delete

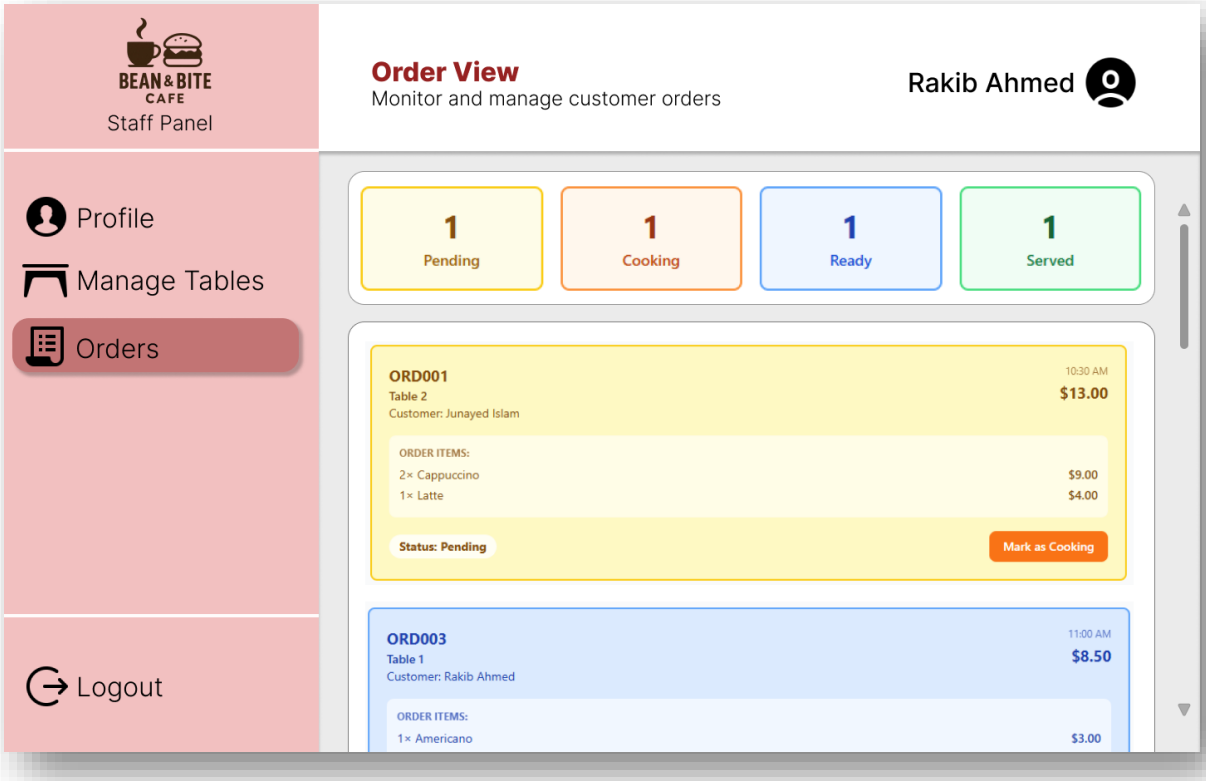

BEAN & BITE
CAFE
Admin Panel
Manage Users
View and manage all registered users
Fahimul Islam
Overview
Products
Manage Users
Financial Reports
Profile
Logout
Total Users
120
Admin
1
Cashier
2
Staff
5
Customers
112
Search by name or email...
All Roles
+ Add New Users
User
Email
Role
Remove User
Fahimul Islam
fahimul@google.com
Admin
Rakib Ahmed
rakib_ahmed@google.com
Staff
MD Farhan
farhan@google.com
Cashier

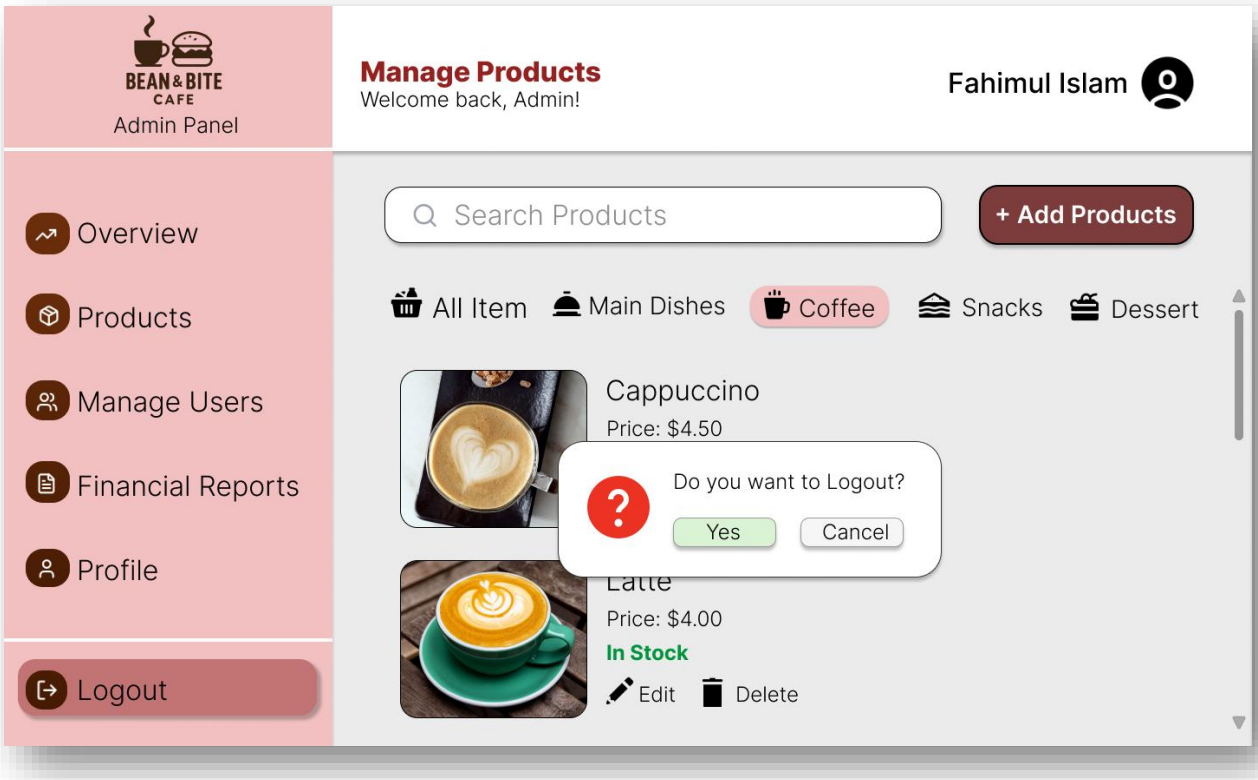

BEAN & BITE
CAFE
Admin Panel
View Profile
Manage your account information
Fahimul Islam
Overview
Products
Manage Users
Financial Reports
Profile
Logout
Fahimul Islam
Admin
Name:
Fahimul Islam
Email Address:
fahimul@google.com
Phone no:
01889877150
Change Password
Current Password:
New Password:
Confirm Password:


BEAN & BITE
CAFE
Cashier Panel
Order Entry
Select products and manage orders
MD Farhan
Profile
POS
Transactions
Reports
Logout
All Item
Coffee
Main Dishes
Dessert
Snacks
Cappuccino
Price: $4.50
In Stock
3
Add to Cart
Latte
Price: $4.00
In Stock
1
Add to Cart
Cart Items
Cappuccino
$4.50 each
3
$13.50
Total Items: 3
Total Amount:
$13.50
Place Order
Clear Cart

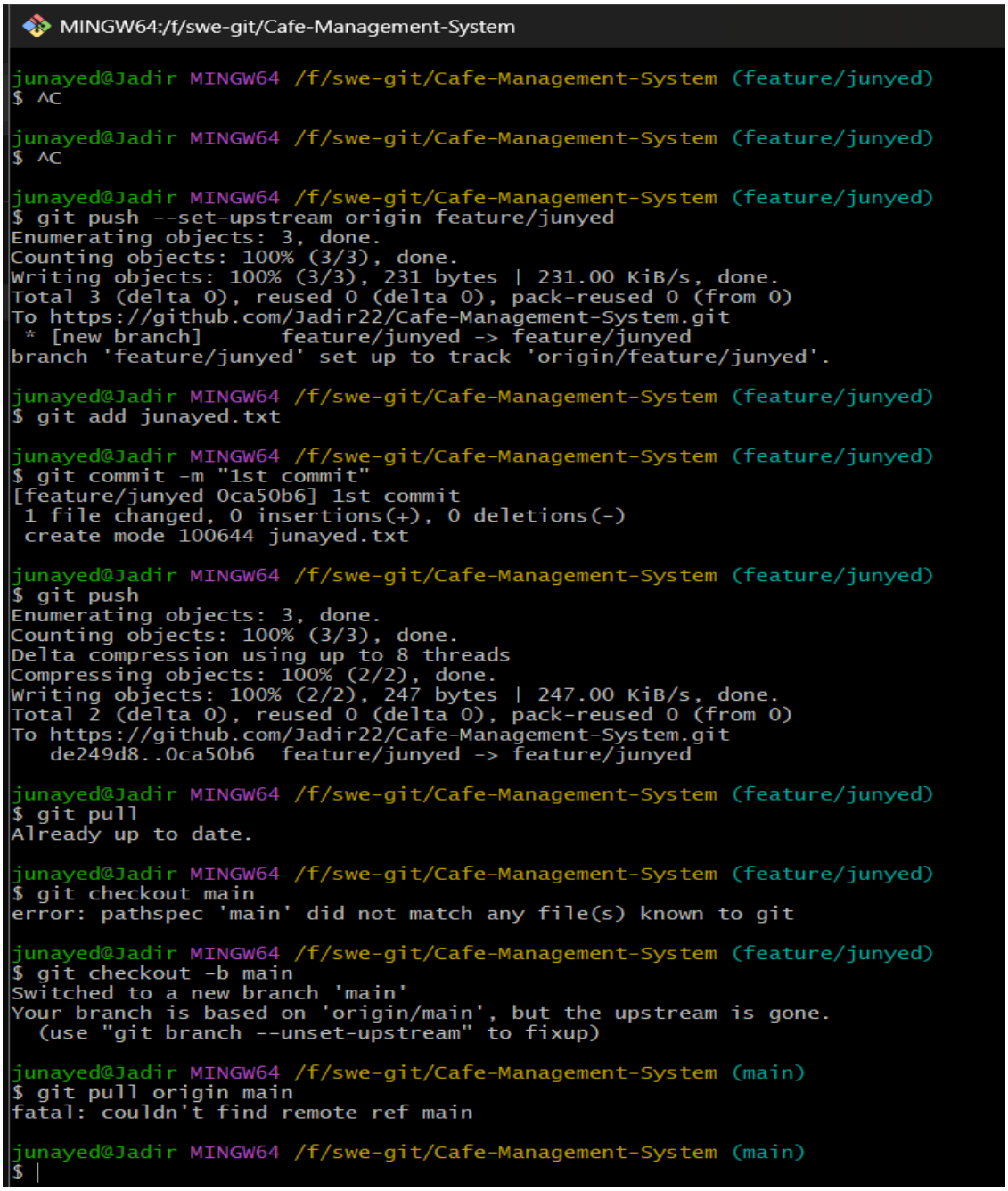

BEAN & BITE
CAFE
Cashier Panel
Payment Process
Complete the transaction and generate receipt
MD Farhan
Profile
POS
Transactions
Reports
Logout
Order Summary
Cappuccino
$4.50 X 3
$13.50
Sub Total:
$13.50
Tax (5%):
$0.675
Total Amount:
$14.175
Payment Details
Select Payment Method
Cash
Card
Process Payment

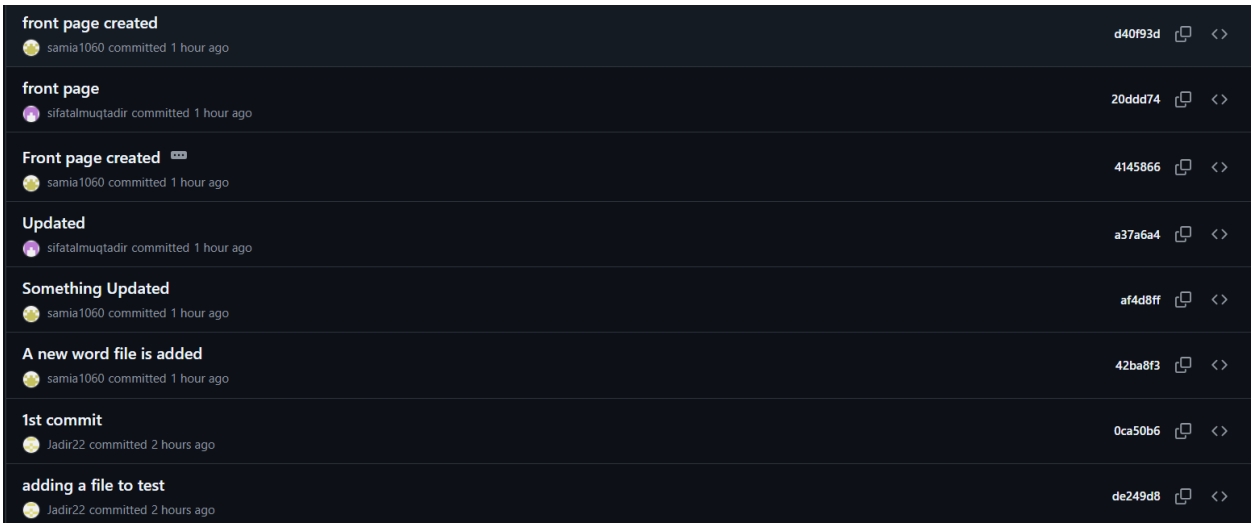

BEAN & BITE CAFE
Cashier Panel
Profile
POS
Transactions
Reports
Logout
Receipt
Generate and print transaction receipt
MD Farhan
BEAN & BITE CAFE
123 Coffee Street, Dhaka
Tel: +880 1234-567890
Date: 2026-01-16
Time: 10:30 AM
Order ID: TXN001
Customer: Junayed Islam
ITEM
AMOUNT
Cappuccino
$4.50 × 3
$13.50
Sub Total: $13.50
Tax (5%): $0.675
Total Amount: $14.175
Print Receipt
Back to POS

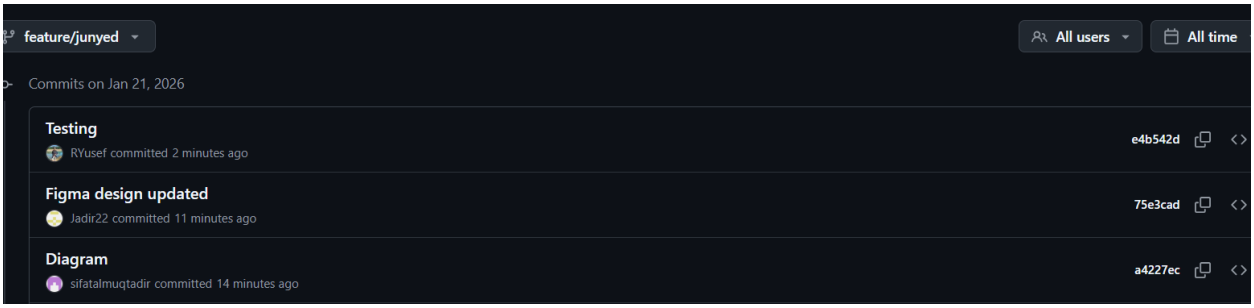

BEAN & BITE CAFE
Staff Panel
Profile
Manage Tables
Orders
Logout
Order View
Monitor and manage customer orders
Rakib Ahmed
1 Pending
1 Cooking
1 Ready
1 Served
ORD001
Table 2
Customer: Junayed Islam
10:30 AM
$13.00
ORDER ITEMS:
2× Cappuccino $9.00
1× Latte $4.00
Status: Pending
Mark as Cooking
ORD003
Table 1
Customer: Rakib Ahmed
11:00 AM
$8.50
ORDER ITEMS:
1× Americano $3.00

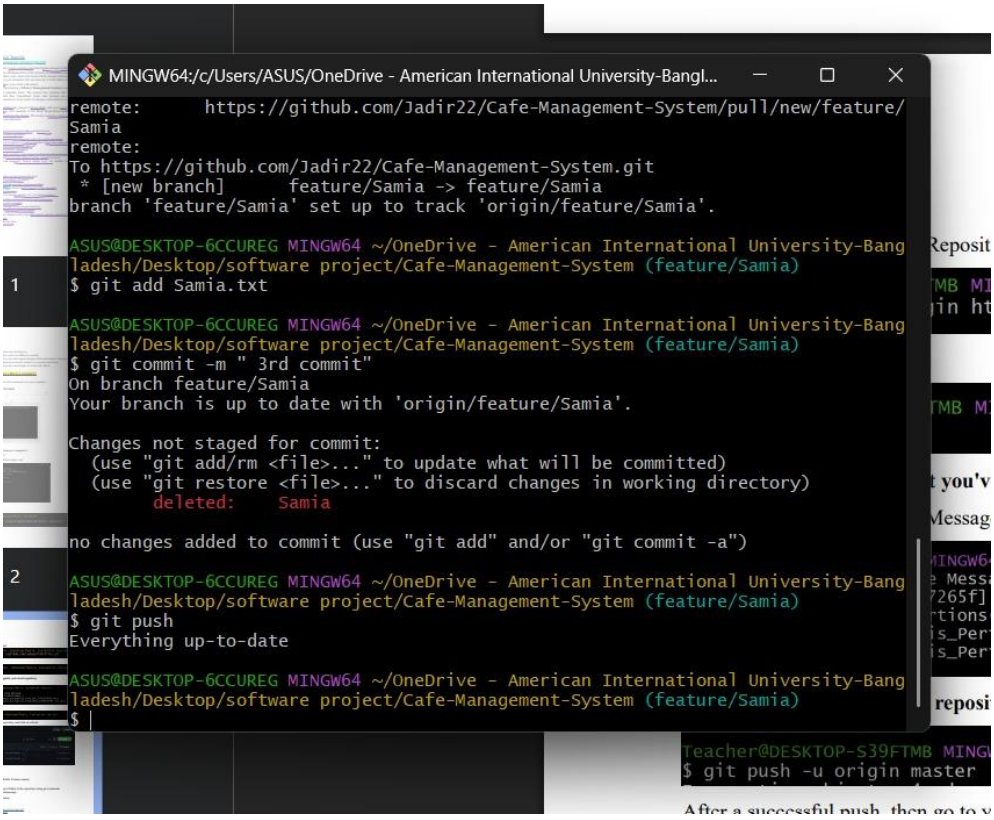

BEAN & BITE
CAFE
Admin Panel
Manage Products
Welcome back, Admin!
Fahimul Islam
Overview
Products
Manage Users
Financial Reports
Profile
Logout
Search Products
+ Add Products
All Item
Main Dishes
Coffee
Snacks
Dessert
Cappuccino
Price: $4.50
Do you want to Logout?
Yes
Cancel
Price: $4.00
In Stock
Edit
Delete

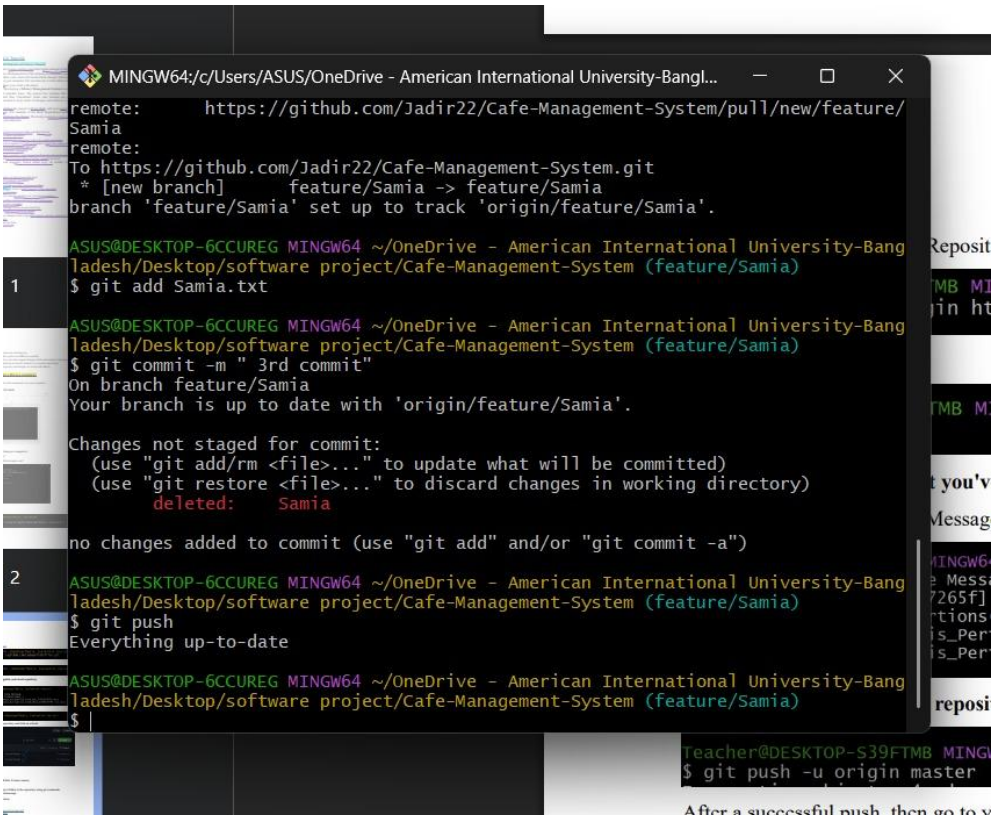
4. GIT WORKFLOW




front page created
samia1060 committed 1 hour ago
d40f93d
front page
sifatalmuqtadir committed 1 hour ago
20ddd74
Front page created
samia1060 committed 1 hour ago
4145866
Updated
sifatalmuqtadir committed 1 hour ago
a37a6a4
Something Updated
samia1060 committed 1 hour ago
af4d8ff
A new word file is added
samia1060 committed 1 hour ago
42ba8f3
1st commit
Jadir22 committed 2 hours ago
0ca50b6
adding a file to test
Jadir22 committed 2 hours ago
de249d8


feature/junyed
All users
All time
Commits on Jan 21, 2026
Testing
RYusef committed 2 minutes ago
e4b542d
Figma design updated
Jadir22 committed 11 minutes ago
75e3cad
Diagram
sifatalmuqtadir committed 14 minutes ago
a4227ec


MINGW64:/c/Users/ASUS/OneDrive - American International University-Bangl...
remote: https://github.com/Jadir22/Cafe-Management-System/pull/new/feature/Samia
remote:
To https://github.com/Jadir22/Cafe-Management-System.git
* [new branch] feature/Samia -> feature/Samia
branch 'feature/Samia' set up to track 'origin/feature/Samia'.
ASUS@DESKTOP-6CCUREG MINGW64 ~/OneDrive - American International University-Bangladesh/Desktop/software project/Cafe-Management-System (feature/Samia)
$ git add Samia.txt
ASUS@DESKTOP-6CCUREG MINGW64 ~/OneDrive - American International University-Bangladesh/Desktop/software project/Cafe-Management-System (feature/Samia)
$ git commit -m " 3rd commit"
On branch feature/Samia
Your branch is up to date with 'origin/feature/Samia'.
Changes not staged for commit:
(use "git add/rm <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
deleted: Samia
no changes added to commit (use "git add" and/or "git commit -a")
ASUS@DESKTOP-6CCUREG MINGW64 ~/OneDrive - American International University-Bangladesh/Desktop/software project/Cafe-Management-System (feature/Samia)
$ git push
Everything up-to-date
ASUS@DESKTOP-6CCUREG MINGW64 ~/OneDrive - American International University-Bangladesh/Desktop/software project/Cafe-Management-System (feature/Samia)
$
1
2
Teacher@DESKTOP-S39FTMB MINGW
$ git push -u origin master


MINGW64:/c/Users/ASUS/OneDrive - American International University-Bangl...
remote: https://github.com/Jadir22/Cafe-Management-System/pull/new/feature/Samia
remote:
To https://github.com/Jadir22/Cafe-Management-System.git
* [new branch] feature/Samia -> feature/Samia
branch 'feature/Samia' set up to track 'origin/feature/Samia'.
ASUS@DESKTOP-6CCUREG MINGW64 ~/OneDrive - American International University-Bangladesh/Desktop/software project/Cafe-Management-System (feature/Samia)
$ git add Samia.txt
ASUS@DESKTOP-6CCUREG MINGW64 ~/OneDrive - American International University-Bangladesh/Desktop/software project/Cafe-Management-System (feature/Samia)
$ git commit -m " 3rd commit"
On branch feature/Samia
Your branch is up to date with 'origin/feature/Samia'.
Changes not staged for commit:
(use "git add/rm <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
deleted: Samia
no changes added to commit (use "git add" and/or "git commit -a")
ASUS@DESKTOP-6CCUREG MINGW64 ~/OneDrive - American International University-Bangladesh/Desktop/software project/Cafe-Management-System (feature/Samia)
$ git push
Everything up-to-date
ASUS@DESKTOP-6CCUREG MINGW64 ~/OneDrive - American International University-Bangladesh/Desktop/software project/Cafe-Management-System (feature/Samia)
$
1
2
Teacher@DESKTOP-S39FTMB MINGW
$ git push -u origin master

```
TUF Raiyan@TUF-Raiyan MINGW64 /d/University Books/Software/Project/Github/Cafe-M
anagement-System (main)
$ git checkout -b feature/Raiyan
Switched to a new branch 'feature/Raiyan'

TUF Raiyan@TUF-Raiyan MINGW64 /d/University Books/Software/Project/Github/Cafe-M
anagement-System (feature/Raiyan)
$ echo "## Task by Raiyan: Implementation of PRD, NFR, Use Case Diagram, User St
ory and Figma" >> PROJECT_FEATURES.md

TUF Raiyan@TUF-Raiyan MINGW64 /d/University Books/Software/Project/Github/Cafe-M
anagement-System (feature/Raiyan)
$ echo "This task involves creating the Product Requirements Document (PRD) detailing the system featur
es, specifications, flow, and acceptance criteria." >> PROJECT_FEATURES.md

TUF Raiyan@TUF-Raiyan MINGW64 /d/University Books/Software/Project/Github/Cafe-Management-System (featu
re/Raiyan)
$ echo "This task involves defining Non-Functional Requirements (NFR) for the system, covering performa
nce, security, usability, and reliability aspects." >> PROJECT_FEATURES.md

TUF Raiyan@TUF-Raiyan MINGW64 /d/University Books/Software/Project/Github/Cafe-Management-System (featu
re/Raiyan)
$ echo "This task involves designing a Use Case Diagram for the system, showing actors and their intera
ctions with various functionalities." >> PROJECT_FEATURES.md

TUF Raiyan@TUF-Raiyan MINGW64 /d/University Books/Software/Project/Github/Cafe-Management-System (featu
re/Raiyan)
$ echo "This task involves creating User Stories for the system, describing end-user goals, acceptance
criteria, and scenarios for each functionality." >> PROJECT_FEATURES.md

TUF Raiyan@TUF-Raiyan MINGW64 /d/University Books/Software/Project/Github/Cafe-Management-System (featu
re/Raiyan)
$ echo "This task involves creating Figma design for the Generate Receipt screen, including layout, but
tons, and interaction flow." >> PROJECT_FEATURES.md

TUF Raiyan@TUF-Raiyan MINGW64 /d/University Books/Software/Project/Github/Cafe-Management-System (featu
re/Raiyan)
$ echo "This task involves creating Figma design for the Login screen, including email/password fields,
 buttons, and validation interactions." >> PROJECT_FEATURES.md

TUF Raiyan@TUF-Raiyan MINGW64 /d/University Books/Software/Project/Github/Cafe-Management-System (featu
re/Raiyan)
$ echo "This task involves creating Figma design for the Logout functionality, showing confirmation flo
w and navigation back to Login or Home screen." >> PROJECT_FEATURES.md

TUF Raiyan@TUF-Raiyan MINGW64 /d/University Books/Software/Project/Github/Cafe-Management-System (featu
re/Raiyan)
$ echo "*Status: Implemented*" >> PROJECT_FEATURES.md

TUF Raiyan@TUF-Raiyan MINGW64 /d/University Books/Software/Project/Github/Cafe-Management-System (featu
re/Raiyan)
$ echo "" >> PROJECT_FEATURES.md
```

5. SOFTWARE TESTING

Project Name: Café Management System			Test Designed By: Junayed Islam	
Test Case ID: 1			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High):			Test Executed By: Junayed Islam	
Module Name: Signup/Register Session			Test Execution Date: 06/01/2026	
Test Title: Verify successful registration with valid unique data				
Description: Ensure a user can register when all data is valid and unique.				
Precondition: High				
Dependencies: No				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Go to Signup page 2. Enter unique Email/Phone 3. Enter password (8+ chars) 4. Enter matching Confirm Password 5. Click Register	**Email:** newuser@test.com **Pass:** Password123 **Confirm:** Password123	Account created successfully; System redirects to Login page.	As expected	Pass

Project Name: Café Management System			Test Designed By: Samia Khan	
Test Case ID:2			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Samia Khan	
Module Name: Signup / Register			Test Execution Date: 06/01/2026	
Test Title: Verify user signup with valid and invalid registration data				
Description: Verifies user registration functionality				
Precondition: User is not registered and is on the signup page.				
Dependencies: User database and validation rules must be active.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Open Signup page 2. Enter unique email/phone 3. Enter password (≥8 characters) 4. Enter matching confirm password 5. Click Register	Email: user@test.com Phone: 017XXXXXXXX Password: test@1234 Confirm Password: test@1234	User registration is successful and user is redirected to login/dashboard page	As expected	Pass

Project Name: Café Management System			Test Designed By: Junayed Islam	
Test Case ID: 3			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Junayed Islam	
Module Name: Forgot Password			Test Execution Date: 06/01/2026	
Test Title: Verify forgot password functionality using OTP				
Description: Verifies password reset using valid email and OTP.				
Precondition: User is registered and on the Forgot Password page.				
Dependencies: User database and OTP service must be active.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Open Forgot Password page 2. Enter registered email 3. Receive OTP 4. Enter valid OTP 5. Enter new password 6. Confirm password 7. Submit	Email: user@test.com OTP: 456789 New Password: newpass@123	OTP is verified, password is updated successfully, and user is redirected to Login page	As expected	Pass

Project Name: Café Management System			Test Designed By: Junayed Islam	
Test Case ID: 4			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Junayed Islam	
Module Name: Admin Dashboard			Test Execution Date: 06/01/2026	
Test Title: Verify Admin Dashboard functionalities				
Description: Verifies admin product, user, report, and profile management.				
Precondition: Admin is logged in with valid credentials.				
Dependencies: Product, user, and report data must exist in the system.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Login as Admin 2. Add, edit, or remove product 3. Manage user roles 4. View financial reports 5. View and edit profile	Product: Latte Price: 220 User Role: Staff Date Range: Jan 2026	Admin can manage products, users, view reports, and update profile successfully	As expected	Pass

Project Name: Café Management System	Test Designed By: Junayed Islam
Test Case ID: 18	Test Designed Date:06/01/2026
Test Priority (Low, Medium, High): High	Test Executed By: Junayed Islam
Module Name: Forgot Password / Security	Test Execution Date: 06/01/2026
Test Title: Verify password mismatch validation during reset	
Description: Ensure the system prevents password reset if 'New Password' and 'Confirm Password' do not match.	
Precondition: User is on the password reset page after OTP verification.	
Dependencies: OTP service and User database.	

Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Enter New Password 2. Enter different Confirm Password 3. Click 'Update'	**New Password:** Password123 **Confirm Password:** WrongPass456	System should show an error message "Passwords do not match" and block the update.	System accepted the update and redirected to the Login page despite the mismatch.	Fail

Project Name: Café Management System	Test Designed By: Samia Khan
Test Case ID: 5	Test Designed Date:06/01/2026
Test Priority (Low, Medium, High): High	Test Executed By: Samia Khan
Module Name: Add New Product	Test Execution Date: 06/01/2026
Test Title: Verify admin can add a new product with valid data	
Description: Verifies adding a new product with valid details.	
Precondition: Admin is logged in and on the Add Product page.	
Dependencies: Product database and validation rules must be active.	

Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Admin 2. Go to Add New Product page 3. Enter unique product name 4. Enter valid price 5. Click Save	Product Name: Americano Price: 180	Product is added successfully and saved in the database	As expected	Pass

Project Name: Café Management System			Test Designed By: Samia Khan	
Test Case ID: 6			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Samia Khan	
Module Name: Remove Product			Test Execution Date: 06/01/2026	
Test Title: Verify admin can remove an existing product				
Description: Verifies product removal from the system.				
Precondition: Admin is logged in and the product exists in the database.				
Dependencies: Product management module must be active.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Login as Admin 2. Open product list 3. Select existing product 4. Click Remove/Delete 5. Confirm action	Product Name: Americano	Products are removed successfully from the database and no longer appear in the list	As expected	Pass

Project Name: Café Management System			Test Designed By: Samia Khan	
Test Case ID: 19			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Samia Khan	
Module Name: Add New Produc			Test Execution Date: 06/01/2026	
Test Title: Verify price validation for negative values				
Description: Verifies that the system prevents adding a product with a negative price.				
Precondition: Admin is logged in and on the Add Product page.				
Dependencies: Product database and validation rules must be active.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Admin 2. Go to Add New Product page 3. Enter product name 4. Enter a negative price	**Product Name:** Espresso **Price:** -50	System should display an error: "Price cannot be negative"	System saved the product with a negative price and	Fail

5. Click Save		and block the save action.	added it to the product list.	

Project Name: Café Management System			Test Designed By: Samia Khan	
Test Case ID: 7			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Samia Khan	
Module Name: Manage Users			Test Execution Date: 06/01/2026	
Test Title: Verify admin user management functionality				
Description: Verifies admin can view and manage system users.				
Precondition: Admin is logged in with valid credentials.				
Dependencies: User management module and user database must be active.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Admin 2. Open Manage Users page 3. View registered users 4. Click Add New User 5. Select role and create user 6. Attempt to delete own account	Role: Staff Email: staff@test.com	Admin can view all users, create new users, and cannot delete own account	As expected	Pass

Project Name: Café Management System			Test Designed By: Sifat Al Muqtadir	
Test Case ID: 8			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Sifat Al Muqtadir	
Module Name: View & Edit Profile			Test Execution Date: 06/01/2026	
Test Title: Verify view and edit profile functionality for all users				
Description: Test the profile viewing and editing system with role-based restrictions				
Precondition: User is logged in with valid credentials				
Dependencies: User database and profile management module must be active				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Login as user 2. Open View Profile page 3. Verify user details displayed 4. Check Role field	Username: user@test.com Password: user123 Name: John Doe	User details match logged-in session, Role field is read-only, Edit Profile redirects to edit	As expected	Pass

5. Click Edit Profile button 6. Update profile information 7. Save changes	Email: john@test.com Role: Cashier (read-only)	page, profile updates successfully.		

Project Name: Café Management System			Test Designed By: Sifat Al Muqtadir	
Test Case ID: 9			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Sifat Al Muqtadir	
Module Name: Order Entry			Test Execution Date: 06/01/2026	
Test Title: Verify product selection, cart update, and order placement				
Description: Checks adding products, updating cart, stock check, and order placement.				
Precondition: Cashier is logged in; products exist.				
Dependencies : Product and stock data must exist.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Login as Cashier 2. Click on a product 3. View Cart list 4. Increase product quantity 5. Check Total Amount 6. Add product with low stock 7. Click Place Order	Username: cashier@test.com Password: cashier123 Product: Coffee - 150 BDT Quantity: 1 → 3 Product: Cake (Stock: 0)	Product added to cart, quantity updates, total recalculates, stock validated, unique Order ID generated	As expected	Pass

Project Name: Café Management System		Test Designed By: Raiyan Yusuf		
Test Case ID: 10		Test Designed Date: 06/01/2026		
Test Priority (Low, Medium, High): Medium		Test Executed By: Raiyan Yusuf		
Module Name: Customer Dashboard		Test Execution Date: 06/01/2026		
Test Title: Verify Order with select, cart and place order				
Description: Verify Customer Dashboard functionalities				
Precondition: Customer is logged in with valid credentials.				
Dependencies: Menu items and order management system must be available.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Login as Customer 2. View profile details 3. Browse menu 4. Select items and place order 5. Track current order status 6. View past receipts 7. Re-order from receipt	Item: Cappuccino Quantity: 2 Order ID: 1080	Customer can place order, track food status, view previous receipts, and successfully re-order	As expected	Pass

Project Name: Café Management System		Test Designed By: Sharin Binte Shafiq		
Test Case ID: 12		Test Designed Date:06/01/2026		
Test Priority (Low, Medium, High): High		Test Executed By: Sharin Binte Shafiq		
Module Name: Cashier Dashboard		Test Execution Date: 18/01/2026		
Test Title: Verify Cashier Dashboard functionalities.				
Description: Verifies that the cashier can access the dashboard, launch POS, view past transactions, generate end-of-day reports, and manage profile information.				
Precondition: Cashier user is logged in with valid credentials.				
Dependencies: User authentication module and transaction data must exist in the system.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Login as Cashier. 2. Open Cashier Dashboard. 3. Click on Launch POS. 4. View past transactions. 5. Generate end-of-day closing report. 6. View profile. 7. Edit profile and save	Cashier ID: C102 Date: 18/01/2026 Phone: 017XXXXXXXX	Cashier dashboard loads successfully, POS launches correctly, transaction history and closing report are displayed, and profile updates are saved.	As expected.	Pass.

Project Name: Café Management System		Test Designed By: Sharin Binte Shafiq		
Test Case ID: 13		Test Designed Date:06/01/2026		
Test Priority (Low, Medium, High): High		Test Executed By: Sharin Binte Shafiq		
Module Name: Payment process.		Test Execution Date: 18/01/2026		
Test Title: Verify payment process by cashier.				
Description: Verifies that the cashier can process payment and calculate change correctly.				
Precondition: An order has been placed and total amount is displayed.				
Dependencies: Order entry module must be completed successfully.				
Test Steps	Test Data	Expected Results	Actual Results	Status (Pass/Fail)
1. Select placed order. 2. Enter cash received amount. 3. Confirm payment.	Total Amount: 350 Cash Received: 500.	System accepts payment, calculates balance correctly, and confirms successful payment.	As expected.	Pass.

Project Name: Café Management System		Test Designed By: Sharin Binte Shafiq		
Test Case ID: 14		Test Designed Date:06/01/2026		
Test Priority (Low, Medium, High): High		Test Executed By: Sharin Binte Shafiq		
Module Name: Generate Receipt.		Test Execution Date: 18/01/2026		
Test Title: Verify receipt generation by cashier.				
Description: Verifies that the cashier can generate and print receipt after successful payment.				
Precondition: Payment has been completed successfully.				
Dependencies: Payment and printer/PDF generation module must work properly.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1.Complete payment. 2. Click on Print Receipt.	Order ID: 2105 Tax Rate: 5%	Receipt is generated containing Date, Time, Order ID, Item details, subtotal, tax, and total amount.	As expected.	Pass.

Project Name: Café Management System			Test Designed By: Raiyan Yusuf	
Test Case ID: 15			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): Medium			Test Executed By: Raiyan Yusuf	
Module Name: Staff Dashboard			Test Execution Date: 06/01/2026	
Test Title: Verify Staff Dashboard functionalities				
Description: Verifies staff order handling, table management, and profile operations.				
Precondition: Staff user is logged in with valid credentials.				
Dependencies : Orders, tables, and staff profile data must exist in the system.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Login as Staff 2. Open Orders section 3. Update order status 4. Mark order as Served 5. Manage table availability 6. View profile 7. Edit profile and save	Order ID: 1025 Status: Served Table No: 4 (Available) Phone: 018XXXXXXXX	Order status updates correctly, table availability changes successfully, and profile information is updated	As expected	Pass

Project Name: Café Management System			Test Designed By: Raiyan Yusuf	
Test Case ID: 16			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Raiyan Yusuf	
Module Name: Order View (Staff)			Test Execution Date: 06/01/2026	
Test Title: Verify staff can view customer orders and update order status				
Description: Verifies instant order display and status update by staff.				
Precondition: Staff user is logged in and at least one order is placed by a customer.				
Dependencies: Customer order placement module must be working properly.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Login as Staff 2. Go to Order View section 3. Observe newly placed customer order 4. Select the order	Order ID: 1056 Initial Status: Pending	Order should appear instantly after customer places it and status should update sequentially to Cooking, Ready, and Served	As expected	Pass

Project Name: Café Management System			Test Designed By: Junayed Islam	
Test Case ID:17			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Junayed Islam	
Module Name: Logout			Test Execution Date: 06/01/2026	
Test Title: Verify logout functionality for all users				
Description: Test the logout system for session clearing and redirection				
Precondition: User is logged in with valid credentials				
Dependencies: Session management and authentication module must be active				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Login as user 2. Navigate to dashboard 3. Click Logout button 4. Verify session cleared 5. Attempt to access protected page	Username: user@test.com Password: user123 Role: Cashier	Session data cleared, user redirected to login page, cannot access protected pages without re-login	As expected	Pass

Project Name: Café Management System			Test Designed By: Sifat Al Muqtadir	
Test Case ID: 19			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Sifat Al Muqtadir	
Module Name: Order Entry			Test Execution Date: 06/01/2026	
Test Title: Verify stock validation during order placement				
Description: Ensures that items with zero stock cannot be added to an active order.				
Precondition: Cashier is logged in; product exists with 0 stock.				
Dependencies: Product and stock data must exist.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Login as Cashier 2. Select product with zero stock 3. Attempt to add to cart 4. Click Place Order	**Username:** cashier@test.com **Product:** Cake **Stock:** 0	System should prevent adding the item to the cart and display "Out of Stock" warning.	System allowed the out-of-stock "Cake" to be added and successfully generated a unique Order ID.	Fail

Project Name: Café Management System			Test Designed By: Raiyan Yusuf	
Test Case ID: 20			Test Designed Date:06/01/2026	
Test Priority (Low, Medium, High): High			Test Executed By: Raiyan Yusuf	
Module Name: Staff Dashboard (Table Management)			Test Execution Date: 06/01/2026	
Test Title: Verify table status updates automatically after serving				
Description: Verifies that a table's status changes from 'Occupied' to 'Available' once the order is marked as 'Served'.				
Precondition: **:** Staff user is logged in; Table No: 4 is currently marked as 'Occupied'.				
Dependencies : Orders and table management modules must be active.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Login as Staff 2. Open Orders section 3. Select active order for Table 4 4. Change status to 'Served' 5. Check Table Management section	**Order ID:** 1105 **Table No:** 4 **Status:** Served	Table No: 4 should automatically update to 'Available' in the system	The order status updated to 'Served', but Table No: 4 remained 'Occupied' in the table list.	Fail

Project Name: Café Management System		Test Designed By: Sharin Binte Shafiq		
Test Case ID: 21		Test Designed Date:06/01/2026		
Test Priority (Low, Medium, High): High		Test Executed By: Sharin Binte Shafiq		
Module Name: Generate Receipt.		Test Execution Date: 18/01/2026		
Test Title: Verify tax calculation accuracy on receipt				
Description: Verifies that the system applies the correct 5% tax rate to the subtotal in the final receipt.				
Precondition: Payment has been completed successfully.				
Dependencies: Payment and printer/PDF generation module must work properly.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Complete payment for an order. 2. Click on Print Receipt. 3. Verify the Tax amount against the Subtotal.	**Order ID:** 2105 **Subtotal:** 400 **Tax Rate:** 5%	The receipt should show a Tax amount of **20** (5% of 400) and a Total of **420**	The receipt displayed a Tax amount of **40** (10%) and a Total of **440**, ignoring the defined 5% rate.	Fail

Project Name: Café Management System		Test Designed By: Sifat Al Muqtadir		
Test Case ID: 11		Test Designed Date: 06/01/2026		
Test Priority (Low, Medium, High): Medium		Test Executed By: Sifat Al Muqtadir		
Module Name: Menu View (Customer)		Test Execution Date: 06/01/2026		
Test Title: Verify customer can view only available menu items				
Description: Verifies that customers can view only those products that are added by the admin and marked as available.				
Precondition: Customer is logged in and admin has added products with availability status.				
Dependencies: Admin product management module must be functional.				
Test Steps	Test Data	Expected Results	Actual Results	Status(Pass/Fail)
1. Login as Customer 2. Open Menu View 3. Observe listed products	Product A: Available Product B: Not Available	Only products marked as available by admin are displayed in the menu	As expected	Pass

6. CONCLUSION

In conclusion, the Cafe Management System is designed to address the common operational challenges faced by modern cafes, especially small and medium-sized businesses that still rely on manual processes. By digitizing core activities such as order management, billing, menu control, and user role management, the system significantly improves efficiency, accuracy, and overall service quality. Automation of repetitive tasks reduces human errors, saves time, and ensures that daily operations run smoothly without unnecessary delays.

The system's role-based access for Admin, Staff, Cashier, and Customers ensures security, accountability, and proper workflow management. With centralized data storage, cafe owners gain better visibility into sales, performance, and operational records, enabling informed decision-making and long-term business growth. Additionally, the user-friendly interface makes the system easy to learn and use, even for staff with limited technical knowledge.

Overall, this project demonstrates how a well-structured desktop-based software solution can modernize cafe operations, enhance customer satisfaction through faster service, and simplify management tasks. The Cafe Management System not only improves daily workflow efficiency but also provides a scalable foundation that can be enhanced in the future with features such as inventory tracking, reporting, and analytics to further support business success.